الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري
Arab Academy for Science, Technology & Maritime Transport

كلية الحاسبات وتكنولوجيا المعلومات (جنوب الوادي)

College of Computing and Information Technology (South Valley)

B. Sc. Final Year Project

(CS402 PROJECT-II)

الكشف عن أمراض النبات باستخدام التعلم العميق المبني على الذكاء الاصطناعي القابل للتفسير

Plant disease detection using deep learning based on XAI

Presented By:

Abdallah Ashraf | *Soliman Osama*

Mohamed Adel | *Othman Abdulrahim*

Amr Mamdouh | *Ziad Mohamed*

Supervised By:

Dr: Tamer Abdel Latif Ali

Aswan, EGYPT

(2025)

DECLARATION

I hereby certify that this report, which I now submit for assessment on the programme of study leading to the award of Bachelor of Science in Computing and Information Technology, is all my own work and contains no Plagiarism. By submitting this report, I agree to the following terms:

Any text, diagrams or other material copied from other sources (including, but not limited to, books, journals, and the internet) have been clearly acknowledged and cited followed by the reference number used; either in the text or in a footnote/endnote. The details of the used references that are listed at the end of the report are confirming to the referencing style dictated by the final year project template and are, to my knowledge, accurate and complete.

I have read the sections on referencing and plagiarism in the final year project template. I understand that plagiarism can lead to a reduced or fail grade, in serious cases, for the Graduation Project courses.

Student1 Name: Abdallah Ashraf
Registration Number: 211013206
Signed: ___________________________
Date: 01/23/2025

Student2 Name: Soliman Usama
Registration Number: 211013107
Signed: ___________________________
Date: 01/23/2025

Student3 Name: Mohamed Adel
Registration Number: 211013016
Signed: ___________________________
Date: 01/23/2025

Student4 Name: Othman Abdelrahim
Registration Number: 211013286
Signed: ___________________________
Date: 01/23/2025

Student5 Name: Amr Mamdouh
Registration Number: 211013207
Signed: ___________________________
Date: 01/23/2025

Student6 Name: Ziad Mohamed
Registration Number: 211013208
Signed: ___________________________
Date: 01/23/2025

STUDENTS CONTRIBUTION

Plant disease detection using deep learning based on XAI

By:

Abdallah Ashraf

Mohamed Adel

Amr Mamdouh

Soliman Osama

Othman Abdulrahim

Ziad Mohamed

Chapter	Title	Contributors
1	Introduction	Mohamed Adel
2	Literature Review	Soliman Osama
3	Modeling	Ziad Mohamed Abdallah Ashraf
4	Prototyping	Amr Mamdouh Othman Abdulrahim Abdallah Ashraf Mohamed Adel Ziad Mohamed Soliman Usama
5	Conclusions and Future Work	Mohamed Adel
6	References	Mohamed Adel

ACKNOWLEDGMENT

We would like to express our deepest thanks and gratitude to initially Allah then to the Arab Academy for Technology and Science for continued support, as the main sponsor and incubator of innovation and creativity. The Academy plays an important role in nurturing research, honing skills, shaping minds, and building character for the young. Its undeniable impact on the Arab community and commendable contributive to producing high caliber alumni who can serve and meet all the challenges of the technological age, extend our appreciation We would not forget to express deepest and heartfelt thanks to Dr. Tamer Abdel Latif Ali, whose guidance during the research has been of so much value and help. His merits are a sharp pioneer in the Egyptian realm of Information Technology. Feeling privileged to be working with him, we have come across inspiration from his distinctive vision and motivation. Our gratitude extends to him for overseeing the project's implementation, providing us with essential insights and skills. His excellence stands as a source of pride for his significant scientific and humanitarian contributions.

ABSTRACT

This project presents an intelligent system for the detection of plant diseases using deep learning techniques enhanced by Explainable Artificial Intelligence (XAI). The primary aim is to build a reliable and transparent model capable of accurately identifying a wide range of plant diseases while offering clear visual explanations that support decision-making in real agricultural environments. The system was trained on a large-scale dataset consisting of over 100,000 images, covering 39 distinct diseases across various plant categories, including field crops, fruits, vegetables, and industrial crops. The dataset was divided into training, validation, and testing subsets in a 70:15:15 ratio to ensure balanced model evaluation. To optimize performance, the system employed InceptionV3 and MobileNet architectures, assigning the most suitable model to each plant category. The results showed high levels of accuracy, with the InceptionV3 model achieving 98.3% test accuracy for fruits and 94.6% for field crops, while MobileNet achieved 95.8% accuracy for vegetables and 92.9% for industrial crops. Beyond classification performance, the use of XAI techniques such as saliency maps enabled the system to visually highlight the regions of each leaf image that contributed most to the model's prediction, thus increasing user trust and interpretability. Overall, this project provides a practical and scalable solution for early plant disease detection, combining deep learning with transparency to support sustainable and efficient agricultural practices.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ACRONYMS/ABBREVIATIONS

ACRONYM	Full word/phrase of the acronym
CNN	Stands for Convolutional Neural Networks
DP	Stands for Deep Learning
GRA-CAM	Stands for Gradient-weighted Class Activation Mapping
LIME	Stands for Local Interpretable Model-Agnostic Explanations
R-CNN	Stands for Region-based Convolutional Neural Networks
SHAP	Stands for Shapley Additive exPlanations
UI	Stands for User Interface
UX	Stands for User Experience
XGBoost	Stands for Extreme Gradient Boosting
XAI	Stands for Explainable Artificial Intelligence

Chapter One

INTRODUCTION

Efficiency and food security in agriculture are relentless pursuits that must face the silent threats of plant diseases, causing devastation in crops, reducing yields, and inflicting economic hardship on farmers and communities. Most disease-detecting methods are based on manual observation, a process that is time-consuming and prone to human error. Indeed, the need for accurate, timely, and scalable solutions is more critical than ever [1,2].

This documentation highlights our effort in the development of a solution to this pressing issue by leveraging the power of DL and XAI. We introduce a comprehensive system that will change the course of plant disease detection from the conventional manual inspections to an automated data-driven approach [3,4].

This project, envisioned as a final year capstone for a Bachelor of Computing and Information Technology, seeks to explore the synergistic potential of leading-edge technologies in the delivery of practical solutions to real-world agricultural challenges. We do not aim at just finding diseases but to provide farmers with accessible, understandable, and actionable insights to enable timely and effective interventions.

The system we've developed integrates several critical components, designed to function together harmoniously to provide an end-to-end solution for plant health monitoring:[5]

- Deep Learning-Powered Image Analysis: At the heart of our system is a sophisticated Deep Learning (DL) model, leveraging Convolutional Neural Networks (CNNs) to accurately detect diseases through analysing plant images. Our system seeks to automatically detect diseases using image analysis, freeing farmers from the burden of manual inspections [6,7].

- Explainable AI: Realizing the limitation of "black box" AI models, we have integrated techniques of XAI such as LIME and SHAP that generate interpretable explanations for model predictions. This will help in building trust among the users by enabling farmers to understand why a particular disease was diagnosed and make more informed decisions and increase the adoption of the technology [8,9].

- Mobile Application: Included is a user-friendly mobile application, which farmers in the field use to rapidly take and analyse images of their crops to receive real-time diagnosis and recommendations on treatment.

- Sensor Integration: A key feature of our system is the integration of environmental sensors that monitor such vital parameters as temperature, humidity, and soil moisture. This will provide a complete overview of plant health and thus enable predictive analysis for prevention.

- Dashboard Platform: The web-based dashboard developed for all rounded monitoring of all components of the system provides in-depth analytics about the usage, performance metrics, and health regarding a variety of crops. This allows both real-time and long-term supervision over agricultural activities.

Developing this system has taken a journey of continuous research, prototyping, testing, and refinement. This is evidence of commitment and teamwork that characterizes our group. The aim of this document is therefore to comprehensively address the reader regarding the journey from mere idea through to functional prototype. These results, along with their principles, can thus contribute to better prospects for any work in this AI and agricultural sustainability intersection. We don't just want to demonstrate our technical skills through this work but our dedication toward solving practical, very significant challenges in the world [10].

Chapter Two

BACKGROUND AND LITERATURE REVIEW

2.1 Background

Plant diseases have always been one of the main causes of reduced yields, hence becoming a serious threat to food security with an ever-growing global population and consequently an increasing demand for food production. Conventional manual methods of disease detection are often sluggish, inexact, and labor-intensive; therefore, there exists a need to develop efficient and reliable solutions[11,12].

DL, and more precisely CNNs, enables a promising approach to automate the process of detecting diseases by means of image analysis[13].

While these models can achieve high accuracy in the results they give, there have always been some criticisms against the transparency of their nature-a fact that makes “black boxes” stand in the way of user adoption. Added to these are the complexities like data requirements, computational resource needs, and different agricultural settings[14,15].

The integration of XAI into DL models has now been developed to reduce the "black box" effect, thus increasing model adoption with more interpretable predictions[16].

This literature review discusses various research works on the adoption of DL and XAI in plant disease detection, focusing on strengths, weaknesses, and prospects that exist in real-world usage. Since this review will analyze several different approaches, the aim is to outline common trends, best practices, and future directions to maximize the desired impact of this technology in agriculture by way of improving yields, lessening losses, and enhancing food security[17,18].

2.2 Literature Review

2.2.1 CONSTRUCTION OF DEEP LEARNING-BASED DISEASE DETECTION MODEL IN PLANTS

2.2.1.1. Introduction

The purpose of the paper is the description of a disease detection model for plants, which is based on deep learning, and mainly it focuses on crops such as bell peppers, potatoes, and tomatoes. The model exploits Convolutional Neural Networks (CNNs) and the process is carried out through three main steps: crop classification, disease detection, and disease classification. To get the finest result of knowing which plant is which and identifying the disease, the researchers loaded these pre-trained models: ResNet50, GoogLeNet, VGG19, AlexNet, and EfficientNet. Along the line, they also resolved the problem of the “unknown” category, which is referring to crops that are not in the model’s database but providing a more unified model[19].

2.2.1.2. Advantages

- **Precise Results:** This model boasted high accuracy both in crop classification and disease detection where individual models across the board reached more than 99% accuracy in certain situations (i.e., EfficientNet for crop classification and GoogLeNet for disease detection)[19].
- **Step by Step Module:** The three-step process is like an animal’s response to a plant’s health issue and is a simple system that can be easily implemented in the field[19].
- **Usefulness on Smart Agriculture:** A model like this can be incorporated in smart farm systems, which will be able to discover all the diseases without the need for a person to operate the machine[19].
- **Adaptability:** Choosing the best pre-trained model from the analysis of the five used for each step gives certainty of the system performing better with different crops as well as diseases[19].
- **Detection of Unknowns:** The “unknown” category was included as a means of classifying the non-model crops, thus, the “unknown” category helped to avoid the probability of misclassifications when the scale goes up High quality[19].

2.2.1.3. Disadvantages

- Limited Crop Scope: Although the model performs well for the crops tested (bell pepper, potato, tomato), it may not generalize as well to other crops unless they are explicitly included in the training set[19].
- Dependence on Image Quality: The model's performance is highly dependent on the quality and resolution of the images. Low-quality images, especially cropped lesion images, reduced the accuracy[19].
- Manual Data Augmentation: The reliance on data augmentation and preprocessing means that high computational resources are required for model training and optimization[19].

2. 2.1.4. Accuracy

Crop Classification:

- EfficientNet: 99.33%
- GoogleNet: 99.08%
- VGG19: 98.71%

[19]

Disease Detection:

- GoogLeNet for bell pepper: 100%
- VGG19 for potato: 100%
- ResNet50 for tomato: 99.75%

[19]

Disease Classification:

- VGG19 for potato: 99.40%
- EfficientNet for tomato: 97.09%

[19]

Crops:

- Bell Pepper
- Potato
- Tomato
- Non-model crops (Apple, Cherry, Corn, Grape, Peach, Strawberry)

[19]

Diseases:

- Bell pepper:
- Bacterial spot
- Potato: Early blight, Late blight
- Tomato: Bacterial spot, Early blight, Late blight, Tomato mosaic virus

[19]

2.2.1.5. Improvements

Expand the Dataset: Increase the variety of crops and diseases in the training dataset, which could improve the model's applicability across different agricultural settings[19].

Enhance Image Preprocessing: Consider using advanced image enhancement techniques (e.g., noise reduction or contrast adjustments) to improve classification accuracy, especially for images taken in field conditions[19].

Incorporate Active Learning: Allow the model to learn from new, unlabeled data by periodically retraining it with new crop or disease data, improving long-term accuracy[19].

Integrate XAI (Explainable AI): Implement LIME (Local Interpretable Model-agnostic Explanations) or SHAP (Shapley Additive Explanations) to better understand and explain the decision-making process of the model[19].

Use of XAI: XAI can be introduced to interpret and explain the predictions made by the CNN models. In this project, it would help[19]:

1. Visualizing Important Features: XAI techniques like LIME can show which parts of the plant images contributed the most to the model's decision, making it easier to explain why a particular disease or crop was identified[19].

2. Building Trust in Automated Systems: By providing transparent explanations, XAI can increase farmers' trust in the automated disease detection system, showing which symptoms (e.g., lesions, colors) led to the disease diagnosis[19].

3. Error Analysis: XAI could help in identifying common patterns in misclassifications, guiding the model developers on which aspects need improvement[19].

2.2.2 USING DEEP LEARNING FOR IMAGE-BASED PLANT DISEASE DETECTION

2.2.2.1. Introduction

The paper titled “Deep Learning for Plant Disease Detection” was authored by Munaf Mudheher Khalid and Oguz Karan from Altinbas University in Turkey. It was published in 2024 in the International Journal of Mathematics, Statistics, and Computer Science, Volume 2. The paper explores the use of Deep Learning (DL) models, specifically Convolutional Neural Networks (CNNs) and MobileNet architectures, to detect plant diseases, offering an efficient alternative to traditional methods[20].

2.2.2.2. Advantages

High Accuracy and Performance:

The CNN model achieved an accuracy of 89%, with a precision and recall of 96%, and an F1-score of 96%[20].

The MobileNet model demonstrated even higher accuracy at 96%, highlighting the potential of Deep Learning models for precise plant disease detection[20].

Scalability and Efficiency:

The use of automated DL models for disease detection can significantly reduce the time and labor associated with traditional manual methods, making it more scalable for large-scale agricultural applications[20].

Explainable AI (XAI) with GradCAM:

Incorporating GradCAM provides a layer of interpretability by visually illustrating how the model makes decisions, which helps users understand why certain regions of plant images are classified as diseased. This transparency is a significant advantage for building trust in AI systems[20].

Comparative Analysis:

The study compares CNN and MobileNet architectures, allowing a broader understanding of their strengths and weaknesses for specific plant disease detection tasks[20].

Expand the Dataset:

- **Incorporate Diverse Crops and Diseases:** The dataset used in the research is limited to specific crops and diseases. Expanding the dataset to include more plant species and a broader range of diseases would make the models more generalizable[20].
- **Increase Data Volume:** Collecting more images, especially for underrepresented disease classes, will enhance the models' ability to detect diseases with higher accuracy, particularly for those classes where performance was suboptimal (e.g., Tomato_Early_blight) [20].

- **Use High-Resolution Images:** Incorporating higher-quality images with better resolution might help the models detect subtle symptoms of diseases more accurately[20].

2.2.2.3. Disadvantages

Limited Dataset Focus:

The dataset used for the models is based on specific crops and diseases, which limits the generalizability of the models to other types of plants and diseases. Future research would need to expand to include a wider variety of plant species and diseases[20].

Challenges with Certain Disease Classes:

While the models perform well overall, certain classes, like Tomato_Early_blight, show lower precision and recall (CNN precision: 62%, F1-score: 71%). This indicates some issues with misclassifications or failure to detect certain diseases, pointing to a need for further optimization[20].

Computational Resource Requirements:

Despite Mobile Net's focus on efficiency, training and deploying these DL models require significant computational power and infrastructure, which may not be feasible in all agricultural settings, especially in remote or resource-limited regions[20].

No Real-World Deployment:

While the paper suggests the potential for real-time applications, such as integration with IoT or mobile platforms, it does not go into detail about how the model could be practically implemented on a large scale for farmers or other agricultural stakeholders[20].

2.2.2.4. Improvements

Improve Model Architectures

- **Ensemble Learning:** Combine different models, such as CNNs, MobileNet, and other architectures like ResNet or EfficientNet, to create an ensemble that leverages the strengths of each model and potentially improves overall accuracy[20].
- **Transfer Learning with Advanced Models:** Experiment with more advanced pre-trained models like Vision Transformers (ViT) or EfficientNet to see if they outperform MobileNet and CNN in plant disease detection[20].
- **Model Optimization for Mobile Deployment:** Since MobileNet was designed for mobile and embedded systems, further optimization (e.g., model pruning, quantization) can reduce computational complexity, making the models more efficient for real-time applications in farming[20].

Implement Real-World Applications

- **Develop a Mobile App:** Deploy the trained models in a mobile or web application to allow farmers to use the technology directly in the field, where real-time disease detection is crucial. This could also integrate additional features like pest detection, soil health monitoring, and weather forecasts[20].
- **Integrate with IoT:** Implement the models on Internet of Things (IoT) devices, such as drones or autonomous robots, for automated large-scale disease detection across agricultural fields[20].

2.2.3 PLANT DISEASE DETECTION USING AI BASED VGG-16 MODEL

2.2.3.1. Introduction

Farming is essential globally for supplying more food due to the fast growth of the population. Unfortunately, low crop production due to disease leads to a lack of food. Traditional manual methods for detecting plant diseases are not only slow but also prone to mistakes[21].

Therefore, early and precise disease detection is essential. Now, machine learning and deep learning, particularly deep learning, can be used for quick and accurate disease identification[21].

Because they are good at recognizing features, especially the VGG-16, the introduced CNN model is excellent in disease classification. And through 15,915 training images, 19 categories of plant diseases were identified from Plant Village dataset[21].

The VGG-16 plant disease model experiment shows a 95.2% accuracy rate. It can detect diseases with only a 0.4418 loss. This means that deep learning can identify plant diseases rapidly and accurately[21].

Using the model will find diseases early. Farmers can prevent major losses by taking appropriate actions quickly[21].

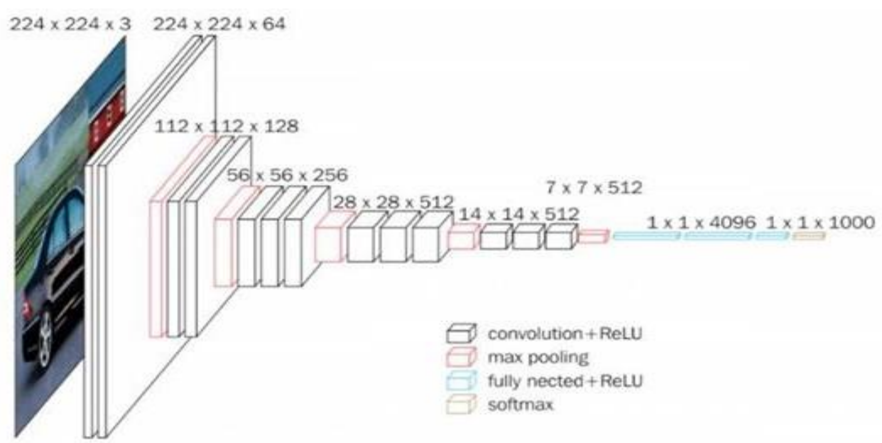

Figure.2.1. VGG-16 Detailed Architecture showing the Various Layer and their Positioning.

[21]

2.2.3.2. Advantages

- **High Accuracy:** Deep learning through VGG-16 software has demonstrated plant disease diagnosis at 95.2% accuracy which is the paper under discussion. This level of accuracy often proves to be a difficult task with the traditional manual inspection or simpler machine learning methodologies[21].

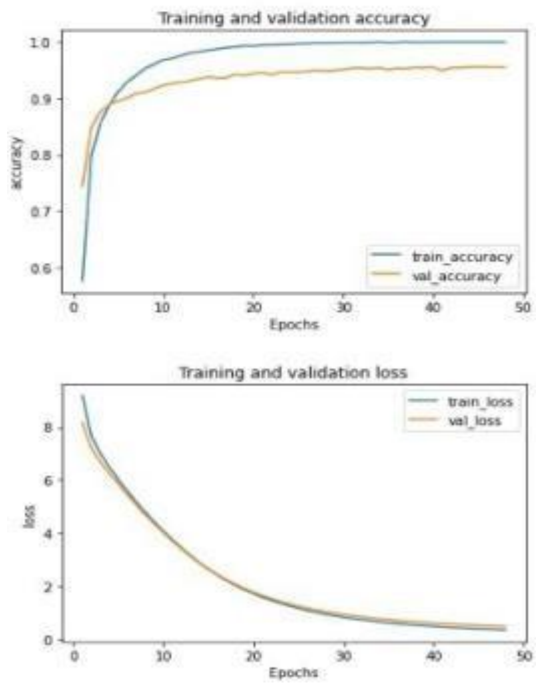

Figure. 2.2. The Performance of Proposed Model in Terms of Training and Validation Loss and Accuracy.

- **Automatic Feature Extraction:** Deep learning models like VGG-16 are especially good at identifying features in an image. That is these kinds of models don't need human feature engineering which takes a while and often requires sustainable knowledge of engineering as it is. The model learns the structure of the same side e.g. by a network training process through the data which in the end is (uniquely) reusable[21].

- **Generalizability:** Deep learning models can usually generalize well to various kinds of plants and diseases even if one cannot name them precisely. In other words, they could learn from the past and be able to predict what the new unseen data could be[21].

- **Time Efficiency:** AI-based disease detection can be much faster than traditional manual inspection and examination. This is vital to the operation of large-scale farms which die or live like time[21].

2.2.3.3. Disadvantages

- Data Requirements: To develop deep learning models, sizeable, labeled data sets are required for data training. This, particularly in difficult or costly times of the year where the data is collected and labeled, can be a real stressor[21].

- Computational Cost: Deep learning methods are very resource-consuming. They consume a lot of powerful hardware and can thus become costly, especially in resource-scarce environments[21].

- Black Box Nature: Deep learning algorithms are often referred to as “black boxes” which has to do with the fact that the procedures leading to decision-making are not easily interpretable. The issue of understanding results of the model and recognizing problems when mistakes happen might be very much difficult[21].
- Sensitivity to Image Quality: It is possible that the deep learning systems can be adversely affected by substandard images. Misleading light, noise, or blurred images can make the model quite ineffective[21].

2.2.3.4. Improvements

- Larger Dataset: The more data you train your model on, the more accurate and generalizable it will become. If possible, try to collect or access a larger dataset encompassing more plant species, diseases, and image variations (lighting, angles, backgrounds) [21].

- **Fine-Tuning Techniques:** Explore advanced fine-tuning techniques to adapt the pre-trained VGG-16 model more effectively to your specific dataset[21].
  This could involve:

  Transfer Learning Strategies: Experiment with different ways to freeze or unfreeze layers in the pre-trained model to optimize learning[21].
  Hyperparameter Optimization: Systematically test different hyperparameters (learning rate, batch size, optimizer, etc.) to find the best settings for your model[21].

  Ensemble Methods: Combine multiple CNN models (e.g., VGG-16 with ResNet or Inception) to create an ensemble. This can often improve performance by averaging predictions[21].

- **Data Augmentation:** This can be done by providing different types of images. Techniques like rotations, flips, zooming, cropping, and color retouching are used to make the software more adaptable to image variations and changing the image resolution[21].

- **Go Beyond Classification to Localization :** Object Detection: Instead of just identifying the type of plant disease your model could also locate the disease, i.e. find its position on the plant leaf. As a result, farmers would be able to know the affected area and therefore they would be able to take suitable action[21].
- **Use Faster : R-CNN or YOLO:** Among the object detection methods, you can think of the Faster R-CNN or YOLO. They are object detection models that are exclusively for that task. Such models can draw a box around the illness spots on the image when they can be used to identify[21].

- **Enhance User Experience :** Interactive Interface: Set up an easy-to-use interface for the farmers where they can simply upload the images, get their predictions, and view their results with clear annotations[21].

- **Mobile App:** Develop a mobile app, where farmers can simply take pictures of their crops while they are on the go and get the results [21].

- Disease Management Recommendations: It must be more than just identifying the diseases thought. Introduce to the user some options of proper therapy, prevention strategies, and the best practices of dealing with diseases that were diagnosed [21].

- **Combine with Other Data Sources:** Sensor Data: Integrate sensor data (temperature, humidity, soil moisture) along with images to create a more comprehensive picture of the plant's health [21].
  Weather Data: Incorporate weather forecasts to predict disease outbreaks and alert farmers to potential risks [21].
- **Focus on Specific Plant Species or Diseases:** Specialized Models: Develop separate models trained on datasets of specific plant species or diseases. This can lead to higher accuracy and better insights into the specific challenges facing particular crops [21].

2.2.4 DEEP LEARNING BASED AUTOMATED DISEASE DETECTION AND CLASSIFICATION MODEL FOR PRECISION AGRICULTURE

2.2.4.1. Introduction

The aim of this research is to study the three most famous rice diseases such as bacterial leaf blight, a disease caused by bacteria, brown spot a disease caused by fungi, and finally, leaf smut which is also caused by fungi. The method described in this paper is going to utilize the Otsu's global thresholding algorithm for the procedure of binarizing images and thus remove any background noise from these images[22].

-Throughout this paper, we have presented a new mechanism for the classification of crop diseases together with the annotation automation of digital photos called the DL-APDDC model. IDEALISM mostly appears in the release of STIS476, which revealed its core intellectual property, aims to categorize the diseased plants that are located in the leaf and fruit areas respectively. The accurate classification of the given technique means it includes various elements: background removal via U2Net Deep Learning, feature extraction by SqueezeNet, classification by Adaboost, and finally, processing via XGBoost. Thus, Fig. 1 diagrammatically describes the whole working of the DL-APDDC model[22].

The Deep Learning-based Automated Plant Disease Detection and Classification (DL-APDDC) model for Precision Agriculture is described in this paper. The developed model is the identification and classification of plant diseases in the leaf and fruit regions[22].

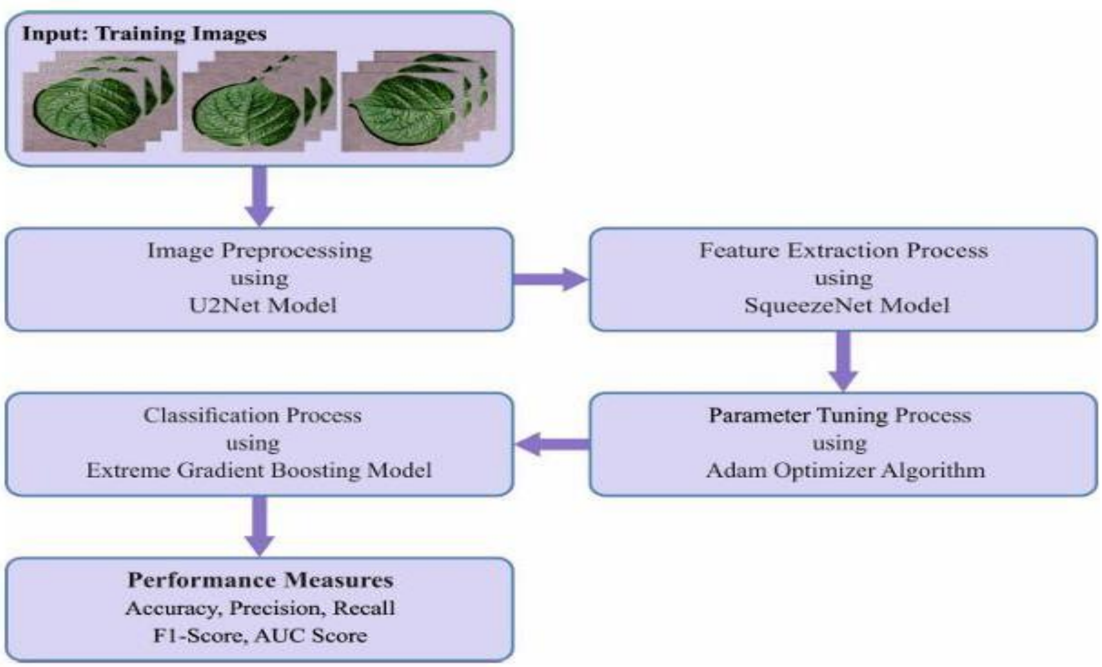

Figure. 2.3. Working process of DL-APDDC system

[22]

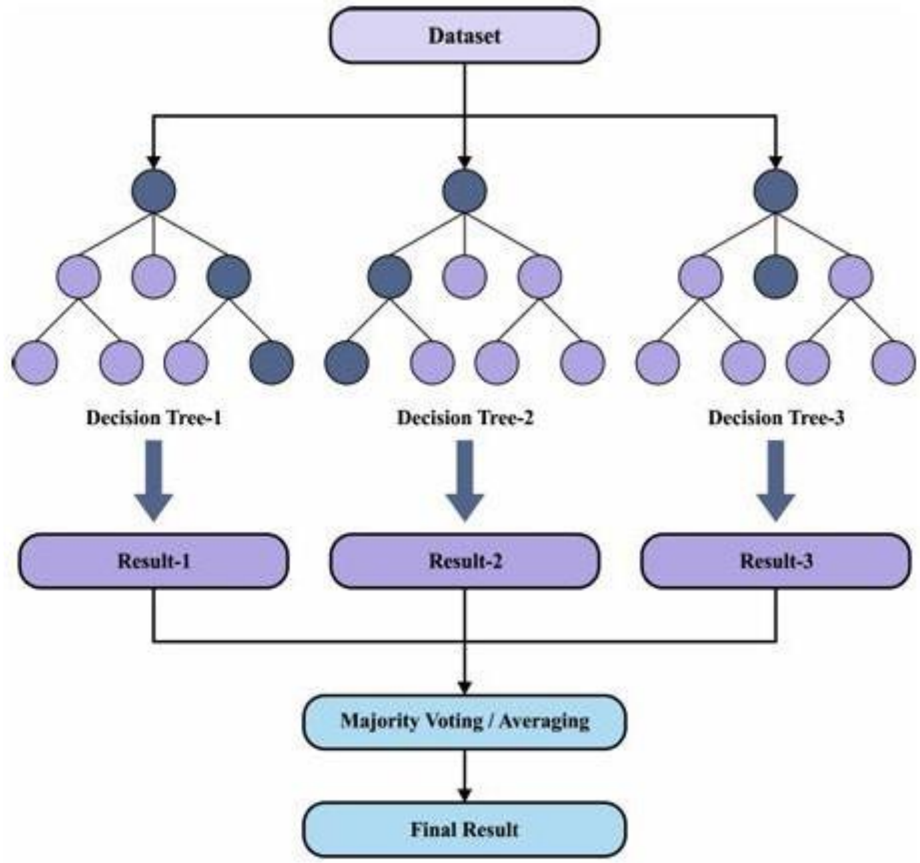

Figure. 2.4. Framework of XGBoost

[22]

2.2.4.2. Advantages

- The DL-APDDC model has achieved high reliability in recognizing agricultural pests with an impressive high of 96.44% and 96.08% accuracy on potato and citrus datasets respectively[22].
- In addition to U2Net-based background removal, the model employs the use of a few advanced techniques such as SqueezeNet feature extraction, Adam optimizer, and XGBoost-based classification which are the contributors of its high-performance level[22].
- The DL-APDDC is a multi-faceted technology deployed on numerous plant diseases hence it is part of modern precision farming[22].

2.2.4.3. Disadvantages

- The paper is silent on the drawbacks of the DL-APDDC device such as its power consumption and the requirement of a high-quality database[22].

- No assessment of comparing the DL-APDDC technology with other state of the art deep learning models for plant disease detection is in the paper[22].

2.2.4.4. Improvements

U2Net-based background removal:

This technique is a process of extracting leaf and fruit regions from input images; therefore, the model will be informed where to focus on in the images[22].

SqueezeNet feature extraction:

This process of obtaining features from the Fruit and leaf images are of course the information that the classifier is going to use to make a prediction[22].

Adam optimizer:

The SqueezeNet model's parameters are the ones determined by the Adam optimizer and it positively influences the model's performance[22].

XGBoost-based classification:

This technology is applied to the classified characteristics of plants by means of the XGBoost method that is a type of machine learning[22].

2.2.4.5. Accuracy

The potato dataset was 96.44% achieved accuracy with the DL-APDDC model, and the citrus dataset was 96.08% achieved accuracy with the same model[22].

Other conventional machine learning methods, such as CNN-SVM, CNN-RF, and CNN-ANN, were outperformed by the model in comparison[22].

The model of DL-APDDC also has a high level of precision, recall, and F1-score on both datasets[22].

To sum it up, the DL-APDDC model can be seen as a good technique for identifying and classifying plant diseases among the tools available in precision agriculture. Further research should be conducted to fulfill the limitations of the model and as well as compare it with other advanced deep learning models[22].

2.2.5 IMPROVING PLANT DISEASE CLASSIFICATION WITH DEEP LEARNING BASED PREDICTION MODEL USING EXPLAINABLE ARTIFICIAL INTELLIGENCE

2.2.5.1. Introduction

Plant ailments are a major source of economic instability, as they are the main reason for less agricultural productivity and affect food safety both on local and global levels. Rapid detection and careful treatment of diseases, at an early stage, are the primary measures that need to be taken to reduce their negative effects. Typically, the diagnosis of diseases has been carried out manually by the experts, which is a time-consuming and subjective process. Recent technological developments, especially in deep learning (DL) have made a huge impact on this field. In this study, a new method of explainable artificial intelligence (XAI)-based deep learning model is used for the classification of plant diseases, which is more accurate and gives a clear view of the model's decision-making process. The essential task is to apply disease recognition while ensuring confidence in the predictions made by the model[23].

2.2.5.2. Technologies Used

- Deep Learning (DL):

  The device that we use for disease diagnosis is EfficientNetB0, the newest deep learning architecture which is mostly known for its efficiency and quality. This model makes use of pre-trained weights to classify 38 different plant diseases. Deep learning techniques such as Convolutional Neural Networks (CNN) enabled the processing of plant disease images, and the identification of patterns to classify them effectively[23].

- Explainable Artificial Intelligence (XAI):

  This Local Interpretable Model-agnostic Explanations (LIME) framework was incorporated into the DL model to clarify the decisions taken by the DL model. LIME is a model that provides feature importance scores, which allows users to know which parts of the input carried most of the model's decisions. This model is a black box, but LIME has managed to make it almost transparent to its users, underlining the positive aspect of the whole development[23].

- Mobile APP:

  A mobile application called 'Plant Care' was made to help farmers identify plant diseases immediately the moment they occur, by classifications using the DL model that was developed[23].

2.2.5.3. Advantages

- High Accuracy:
  The model clinched a remarkable 99.69% accuracy, which was above other state-of-the-art models such as MobileNetV2 and ResNet-50[23].

- Explainability:
  The existence of LIME in the model guarantees that the model's predictions are easy to understand, it gives information about the decision-making process, and it promotes transparency, which is the key to building the trust in automated systems, in particular agriculture applications. Where errors lead to important economic implications[23].

- Practical Application:
  The creation of the mobile app makes it possible for the model to be practically used by farmers for the identification of disease through real-time testing, thus it helps the quick adoption of technology in the agricultural sector[23].

2.2.5.4. Disadvantages

- Dataset Imbalance:
  The dataset that was used in this research was unbalanced, this implies that some diseases had more data of images than others, which the model may find hard to handle for the less likely occurred diseases[23].

- Dependency on Pre-trained Models:
  The study used pre-trained models for feature extraction, which means that the model's adaptability to new or unseen plant diseases, those which were not part of the training dataset, may be restricted[23].

2.2.5.5. Accuracy

The model demonstrated exceptional performance, with the following metrics:

- **Accuracy:** 99.69%
- **Precision:** 98.27%
- **Recall:** 98.26%

2.2.6 EVALUATION OF XAI MODELS FOR INTERPRETATION OF DEEP LEARNING TECHNIQUES' RESULTS IN AUTOMATED PLANT DISEASE DIAGNOSIS

2.2.6.1. Introduction

One of the major problems solved by this research is the accurate diagnosis of plant diseases, which is key for the development of agriculture and food security[24].

Although the use of CNNs has been very high because they not only can find and make predictions about the diseases but also effectively through the images of the plants, they are often criticized for not being able to transparently provide their decision-making process in the form of "black box" models. In this study, XAI models such as LIME, SHAP, and Grad-CAM are used for the interpretation of CNNs in diagnosing plant diseases[24].

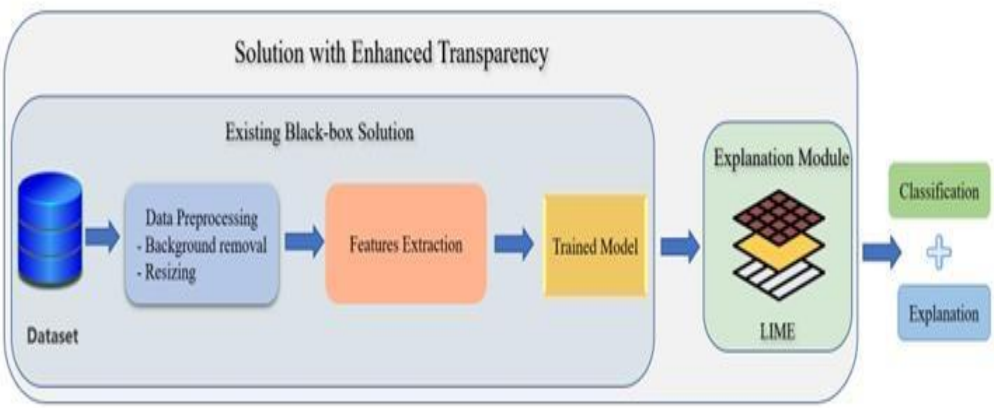

Figure 2.5.: The proposed methodology workflow

[24]

2.2.6.2. Technologies Used

- CNNs (Convolutional Neural Networks): These are very complex learning algorithms that are mostly applied for decision classification in images. However, in our situation CNNs understand the plant disease by examining the leaves[24].
- Explainable Artificial Intelligence (XAI): Techniques such as LIME, SHAP, and Grad-CAM are applied to the model to make output decisions more transparent and interpretable to the user[24].

2.2.6.3. Advantages

- CNNs with the feature of high accuracy: can learn the features that are relevant to the identification of diseases automatically from the images so that the diagnosis of diseases would be more accurate[24].

- Automation: The process can be automated in such a way that there will be minimum human plant inspection, a situation leading to time saving and labor[24].

- Interpretability: The XAI methods deliver the model explanation that helps users to know the causal links that prompt CNN to choose a certain class, which, in turn, results in an increase in their confidences[24].

2.2.6.4. Disadvantages

- Limited Interpretability Without XAI:
  Without XAI, CNNs are difficult to understand, so this could restrict their application to important fields such as agriculture[24].

- Data Requirements:
  CNNs need large, annotated datasets for training, which might be a stumbling block in some agricultural settings. Environmental Dependency: CNN models might behave differently in different situations and as a result, they may be less generalizable[24].

2.2.6.5. Accuracy

The research shows very high accuracy in plant disease detection using CNN models, but specific metrics such as precision and F1-scores are not provided in the summary[24].

The researchers have employed the Plant Village dataset, which contains pictures of healthy plants as well as those with different diseases, and The CNN model, which had the best scores, showed impressively the best results[24].

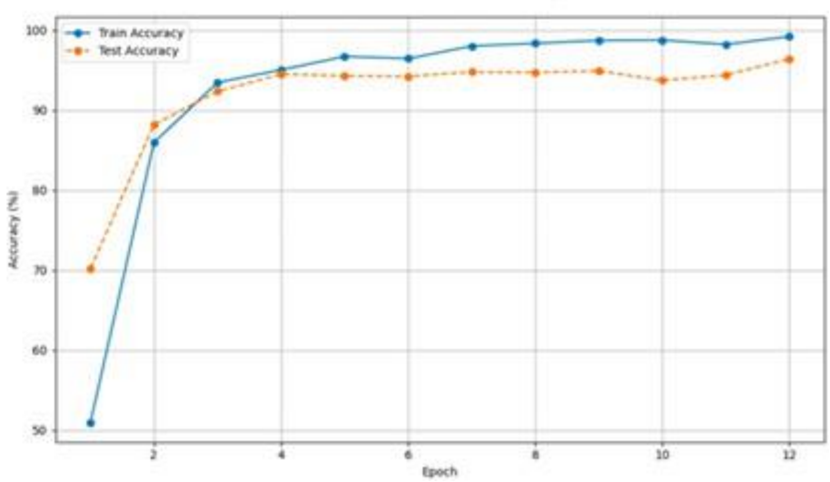

Figure 2.6.: Accuracy Curve for plant disease detection

[24]

2.2.7 Literature Review Comparison Table

	Chapter 1: Deep Learning Model	Chapter 2: CNN & MobileNet	Chapter 3: VGG-16 Model	Chapter 4: DL-APDDC Model	Chapter 5: XAI Enhanced Model	Chapter 6: XAI Evaluation
Focus	3-step disease detection for specific crops	CNN/MobileNet for detection, XAI	Disease classification using VGG-16	Automated detection & classification	XAI to improve disease classification	XAI interpretation of CNN results
Key Methods	Pre-trained CNNs (ResNet, GoogleNet, etc.)	CNN, MobileNet, GradCAM	VGG-16 CNN	U2Net, SqueezeNet, Adaboost, XGBoost	EfficientNet, LIME	CNN, LIME, SHAP, Grad-CAM
Accuracy	99-100% (specific models & crops)	CNN: 89%; MobileNet: 96%	95.2%	96.44% (potato), 96.08% (citrus)	99.69%	High (metrics not detailed)
Specific Crops/Diseases	Bell Pepper, Potato, Tomato, plus some non-model crops.	Specific crops/diseases mentioned	19 plant diseases from Plant Village dataset	Rice diseases (bacterial blight, brown spot, leaf smut), also potato and citrus	38 plant diseases	Various plants and diseases from the PlantVillage dataset
Advantages	High accuracy, step-by-step, adaptability, unknown class	High accuracy, scalability, XAI	High accuracy, automatic feature extraction, efficiency	High accuracy, multi-faceted, uses advanced techniques	High accuracy, explainability, practical app, scalable	High accuracy, automation, interpretability
Disadvantages	Limited crop scope, image quality-dependent, resource intensive	Limited data, some class issues, resources	Limited data, computational cost, "black box," image sensitivity	No discussion on power consumption, no comparison with other DL models.	Dataset imbalance, pre-trained model dependency	Limited without XAI, data requirements, environmental dependency, resource intensive
Improvements	Expand data, better preprocessing, active learning, XAI	Expand data, improve models, optimize, real-world apps	Expand data, fine-tuning, augment data	More diverse crops, comparison with more advanced DL methods	Address data imbalances, train on diverse datasets	Improve transparency, generalize across environments

[1],[2],[3],[4],[5],[6]

C h a p t e r   T h e r e

PROPOSED SYSTEM

3.1 Project Methodology

1. Planning

Objective: Establish a comprehensive roadmap for the project.

Activities:

- Define project goals, scope, and deliverables.
- Allocate resources and create a timeline for development milestones.
- Conduct initial brainstorming sessions to outline system architecture and requirements.

2. Data Collection

Objective: Gather relevant data to support system development and AI modeling.

Activities:

Collect agricultural data on plant health, diseases, and environmental factors.

Research user needs through surveys and interviews with target audiences, such as farmers and gardening enthusiasts.

Compile datasets for AI training and testing.

3. UI/UX Design

Objective: Design user-friendly interfaces for mobile applications and website.

Activities:

Create wireframes and prototypes for the app and website to ensure intuitive navigation.

Focus on accessibility, responsiveness, and aesthetic appeal.

Gather user feedback to refine designs for an optimal user experience.

4. Dataset Preprocessing

Objective: Prepare collected data for use in AI models.

Activities:

Clean and preprocess data to eliminate inconsistencies and enhance quality.

Annotate data for training AI models, such as those for plant disease detection.

Split data into training, validation, and testing subsets.

5. Mobile App Development

Objective: Build a cross-platform mobile application for plant care management.

Activities:

Develop the app using Flutter to ensure a consistent experience across devices.

Implement core features such as AI-based plant care recommendations, a chatbot for real-time assistance, and community interaction tools.

Integrate real-time sensor data visualization and historical trends.

Include push notifications for critical alerts and plant care reminders.

6. Dashboard Development

Objective: Create a centralized dashboard for monitoring and administration.

Activities:

Develop an admin panel for managing app content, user data, and system monitoring.

Display real-time sensor data and analytics for decision-making.

Ensure integration between the dashboard, mobile app, and sensor system.

7. Hardware Development (Sensor System)

Objective: Develop a robust sensor system for real-time plant and environmental monitoring.

Activities:

Assemble and program sensors, including PIR motion sensors, DHT11, moisture sensors, and BME680.

Integrate sensors with ESP8266 NodeMCU for efficient data collection and transmission.

Optimize firmware for energy efficiency and reliable performance.

Test and validate the sensor system in various environmental conditions.

8. Website Development

Objective: Build a comprehensive website to serve as a central hub for the project.

Activities:

Design and develop a responsive website with intuitive navigation.

Include functionality for:

Mobile App Downloads: Host the mobile application for direct user downloads.

Sensor System Purchases: Provide an e-commerce platform to sell the sensor system.

Access to Research Papers: Host project documentation and related research papers for public access.

Test website features for usability and compatibility across devices.

9. Testing and Validation

Objective: Ensure system reliability and user satisfaction.

Activities:

Conduct rigorous unit and integration testing across the mobile app, sensor system, and website.

Validate AI model accuracy in providing plant care recommendations and disease detection.

Collect feedback from beta testers, including farmers and plant care enthusiasts, to refine the system.

Verify data flow and synchronization between components.

10. Deployment

Objective: Launch the integrated system for public use.

Activities:

Deploy the mobile application on the project website for user download.

Launch the e-commerce platform on the website for purchasing the sensor system.

Make the project documentation and research papers available on the website.

Provide user guides and initial technical support to assist early adopters.

11. Project Design & Development

Objective: Design and implement the integrated system, including both the AI-based diagnosis and the embedded hardware component.

Activities:

Design the overall system architecture, including the interaction between mobile app, backend, and embedded device.

Develop the mobile application for capturing plant images and displaying health analysis.

Train and evaluate deep learning models using the prepared datasets.

Implement the embedded system to monitor soil conditions such as pH, moisture, and fertilizer level.

Integrate sensor data with the system dashboard for real-time environmental feedback.

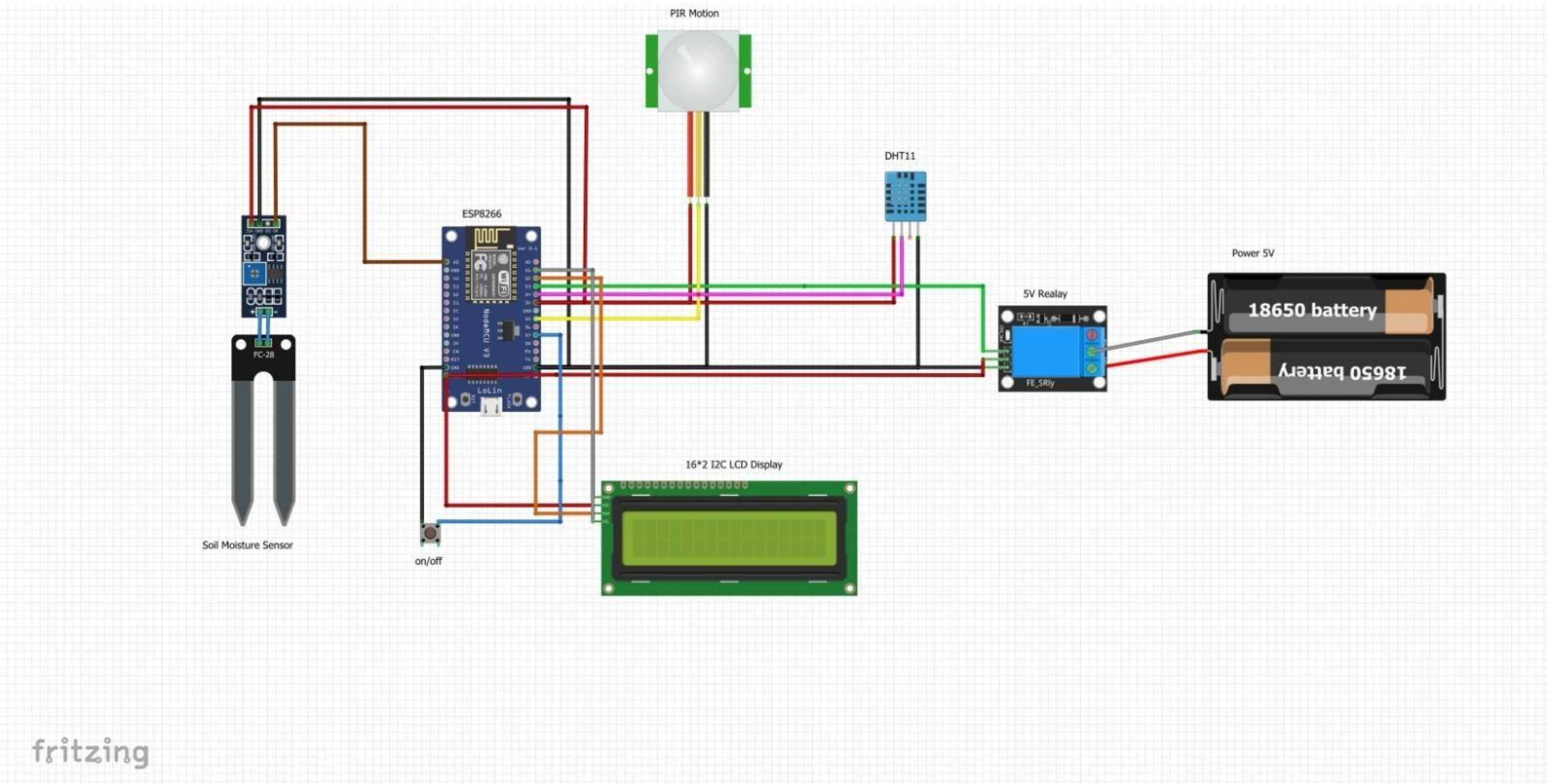

figure 3.1.: Environmental sensor simulation

3.2 Details of Schematic diagram.

DFD (Data Flow Diagram)

Level 0:

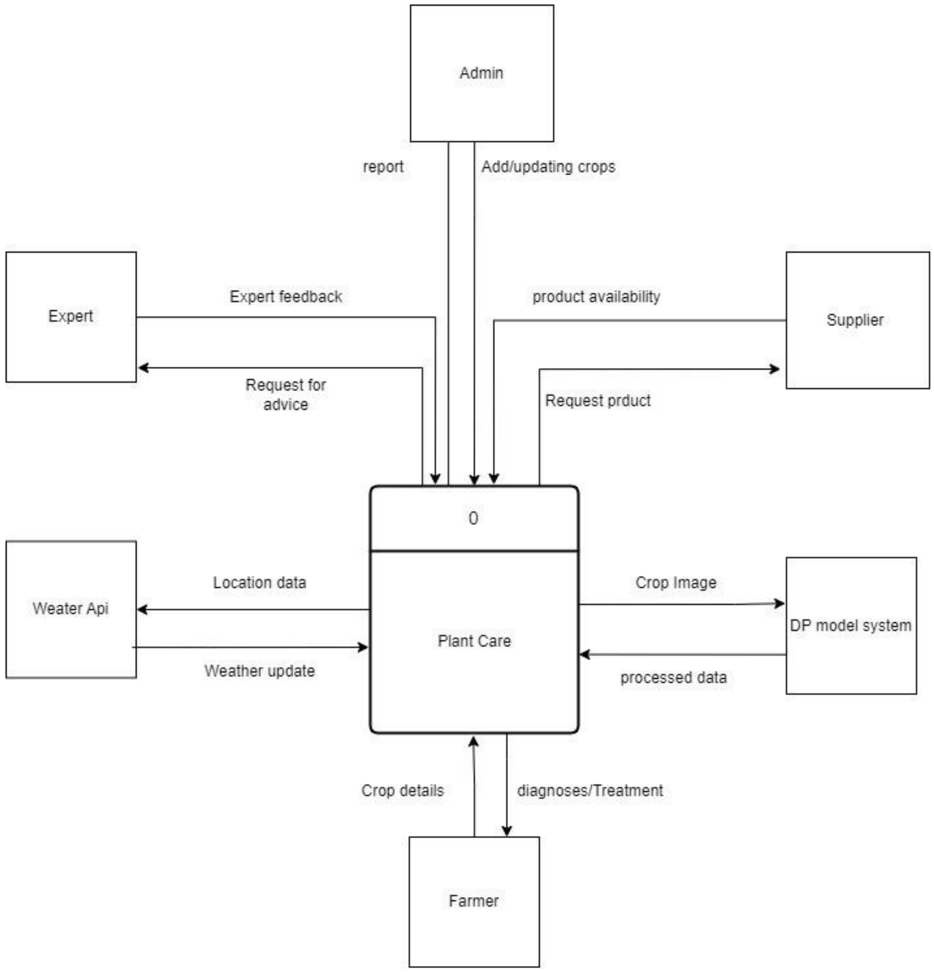

figure 3.2: DFD level 0(Context Level) for proposed system

Level 1:

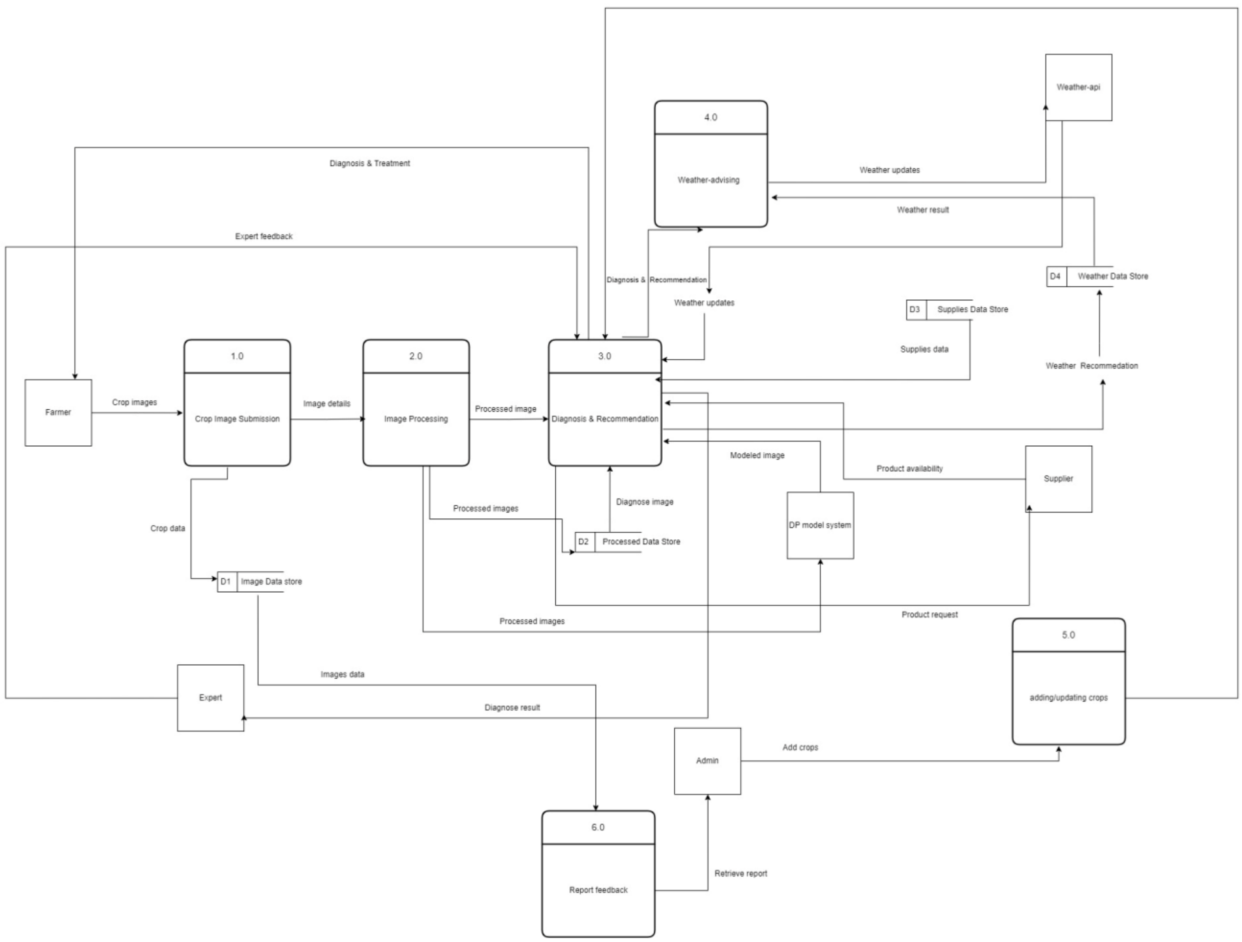

figure 3.3.: DFD level 1(Diagram 0) for proposed system

3.3 Gantt Chart:

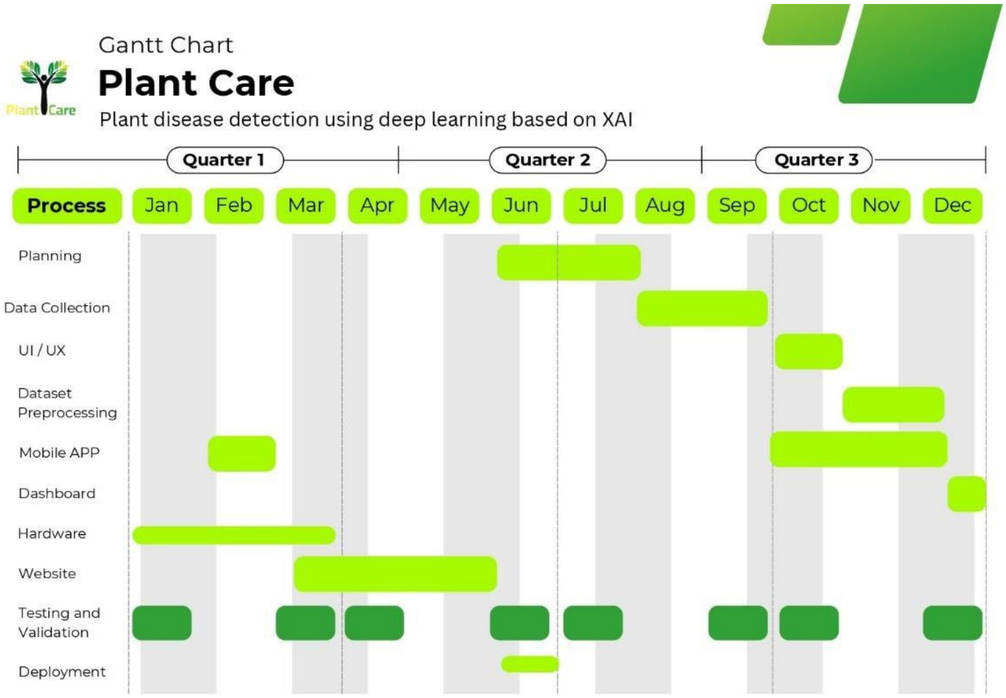

figure 3.4.:Gantt chart

3.4 Business Model:

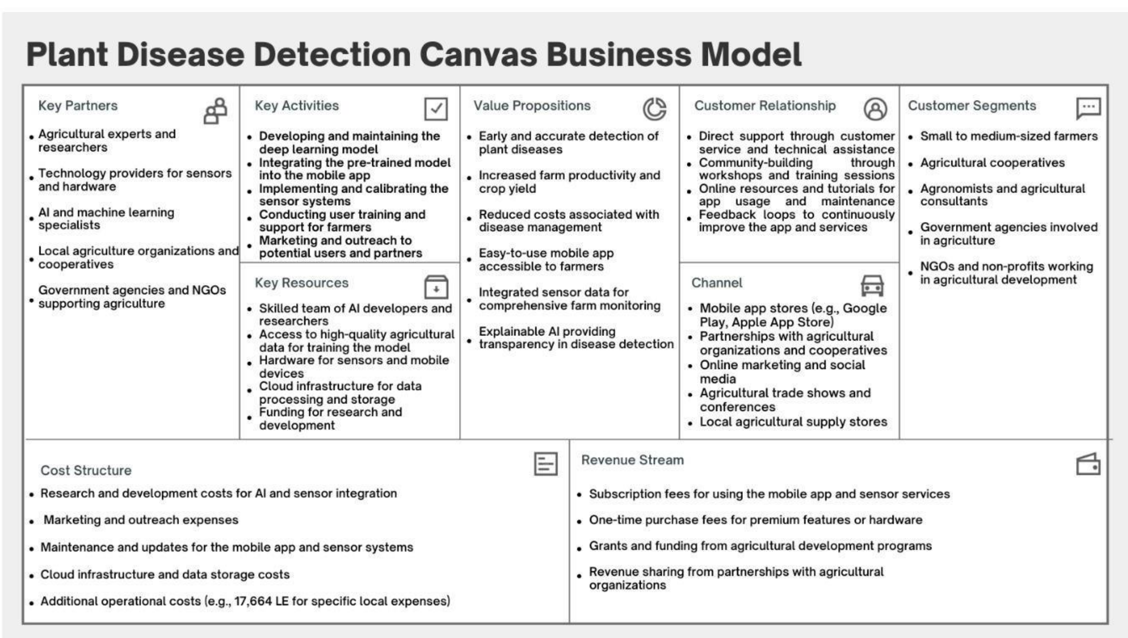

figure 3.5.:Business Model

3.5 Requirements:

- Jam board for Brainstorming
- Figma for UI/UX
- Data
- Python
- TensorFlow with necessary packages
- Segmentation Models
- GPU
- Performance Metrics
- Server to host backend.
- Web framework for Frontend
- Flutter for Mobile apps

3.6 Flow Chart:

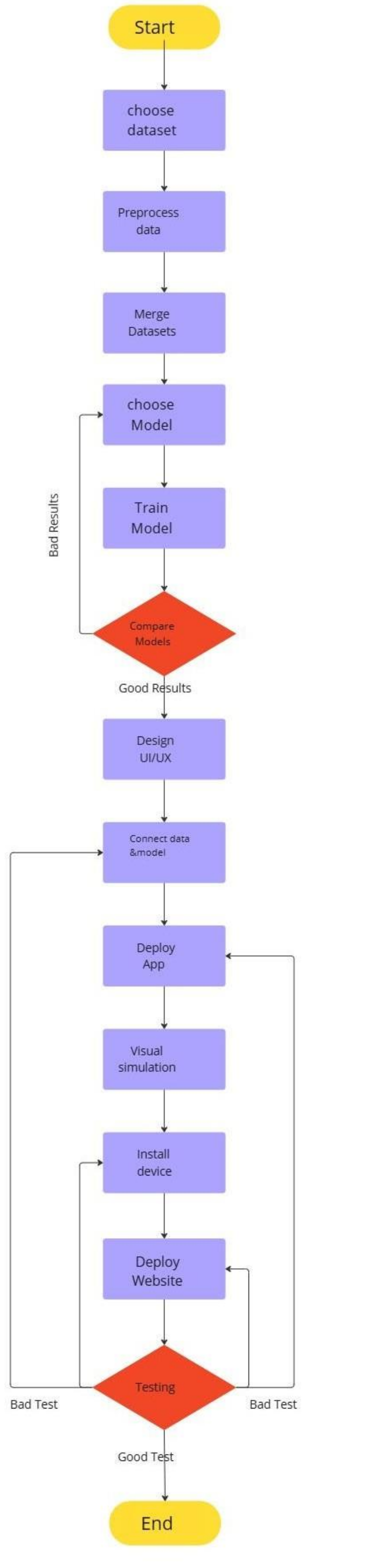

figure 3.6.: Flow Chart

3.7 Mind Map:

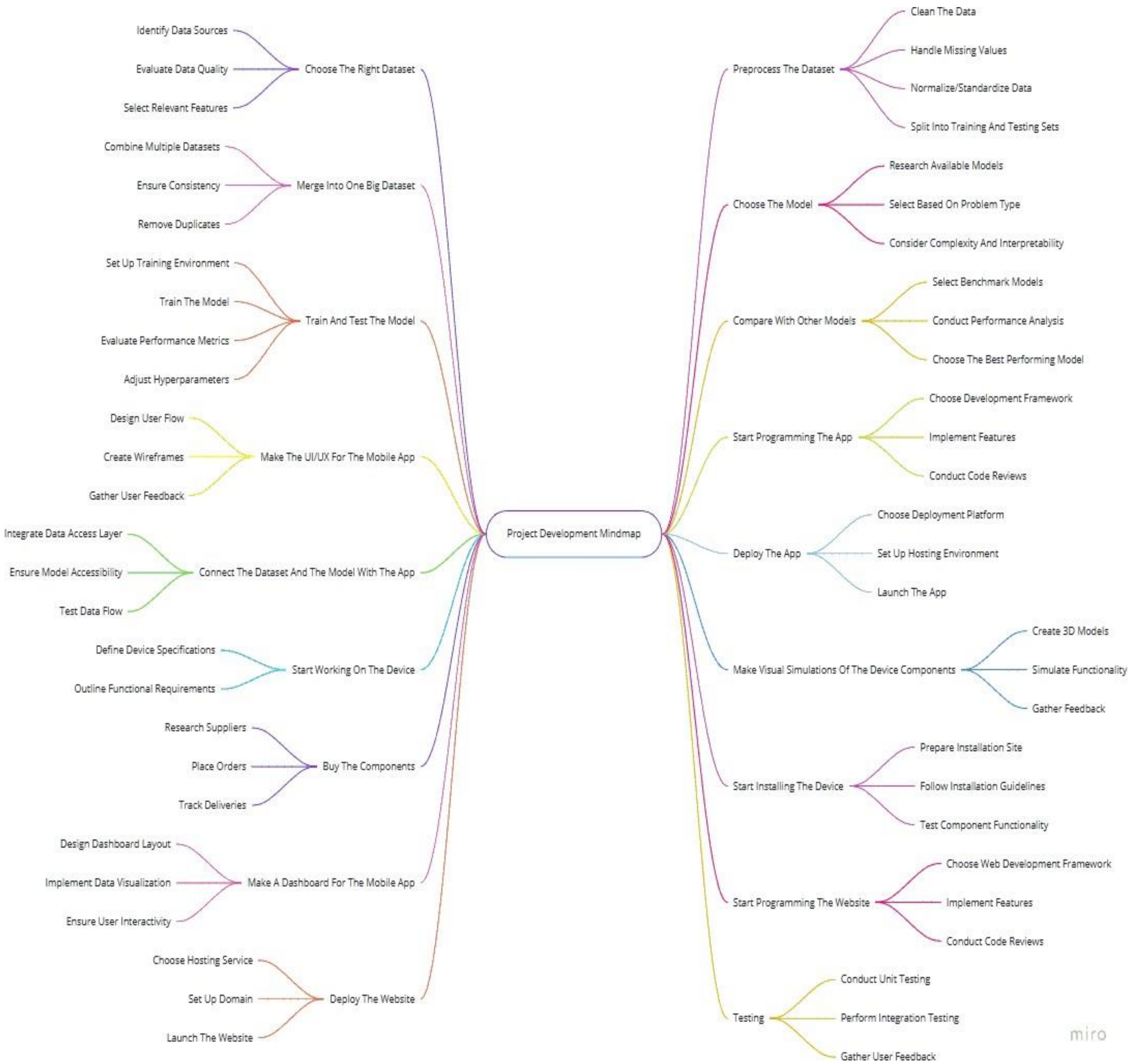

figure 3.7.:Mind Map

RESULTS AND DISCUSSION

4.1 Dataset Overview

figure 4.1:Dataset Crops

- Total Images Before Augmentation:* 100,032

- Number of Diseases Covered:* 39

Dataset Categories:

1- field crops : Corn - Rice - wheat

2- fruits: Grape - strawberry - Citrus

3- industrial crops: cotton – Sugarcane

4- Vegetables: Potato - Tomato

figure 4.2:Dataset Crops

Dataset Splitting:

The dataset was divided into three parts:

- Training Set: 70%

- Validation Set: 15%

- Testing Set: 15%

Reasons for Dataset Division:

1. Efficient Training: Allows the model to focus on specific plant types.

2. Disease Similarity: Diseases within the same plant group often share visual characteristics.

3. Better Performance Analysis: Easier to evaluate model accuracy for each plant type

4. Scalability: Simplifies adding new plant types or diseases.

5. Improved Data Management: Organizing diseases by plant type streamlines data preprocessing.

4.2 RESULTS

4.2.1 field crops:

- **Corn**
- **Rice**
- **Wheat**

IMAGES: 20,922

CLASSES: 3 {9 Disease - 3 healthy}

Model	Validation Accuracy	Test Accuracy
InceptionV3	0.98125	0.94621

This confusion matrix illustrates the performance of the InceptionV3 model on field crops, including corn, rice, and wheat. With a total of 20,922 images spanning 9 diseases and 3 healthy classes, the model achieved a strong test accuracy of 94.62%. The matrix reflects the model's effectiveness in differentiating between disease types and healthy conditions across diverse crop types, as shown in the image below.

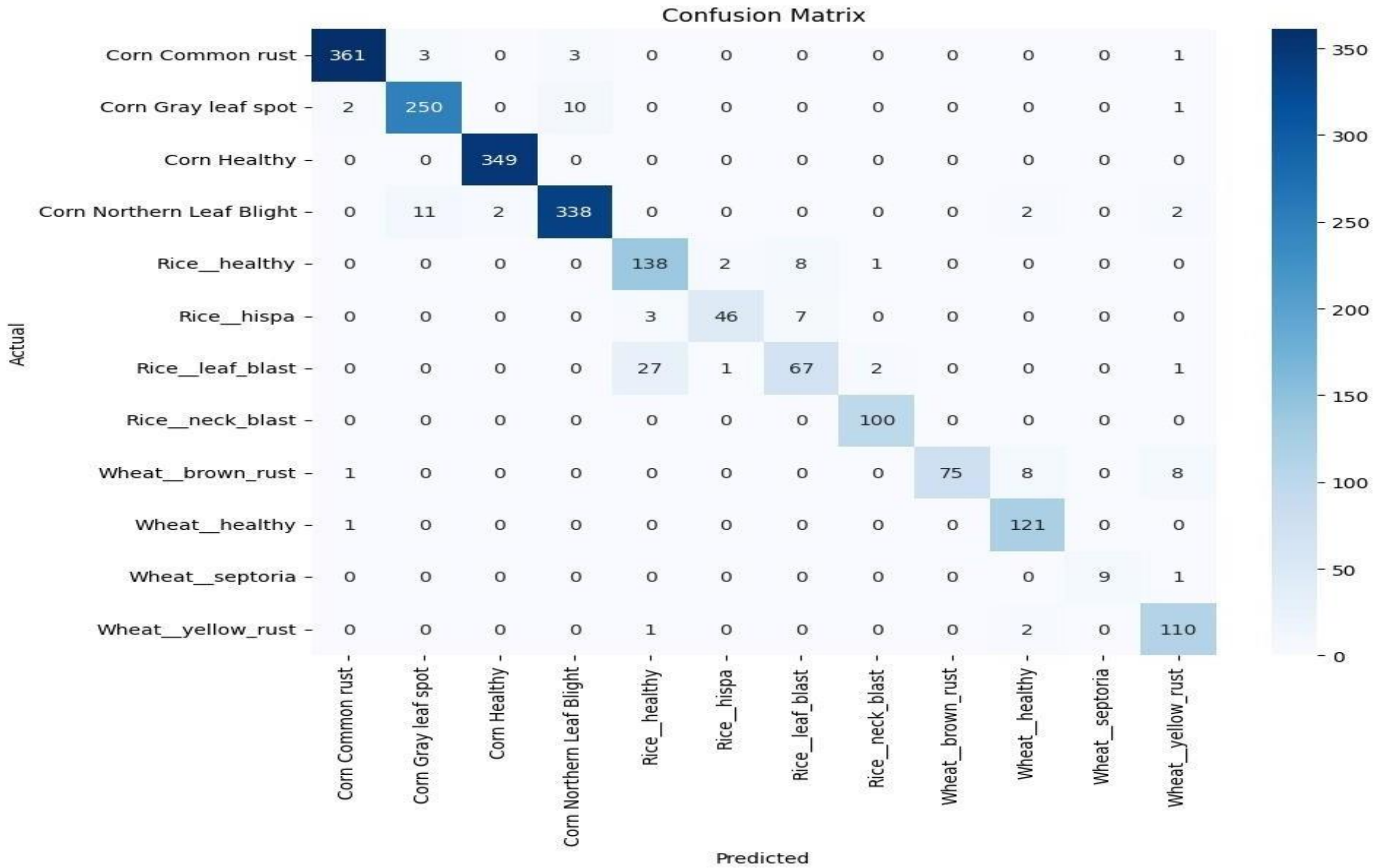

figure 4.3.: Confusion Matrix of InceptionV3

4.2.2 fruits:

- **Grape**
- **strawberry**
- **Citrus**

IMAGES: 26,041

CLASSES: 3 {7 Disease - 3 healthy}

Model	Validation Accuracy	Test Accuracy
InceptionV3	0.98462	0.98301

The confusion matrix below shows the classification results of the InceptionV3 model on fruit crops such as grape, strawberry, and citrus. The dataset contains 26,041 images covering 7 diseases and 3 healthy classes. The model demonstrated excellent performance, achieving a test accuracy of 98.30%, with minimal misclassification between similar disease patterns, as shown in the image below.

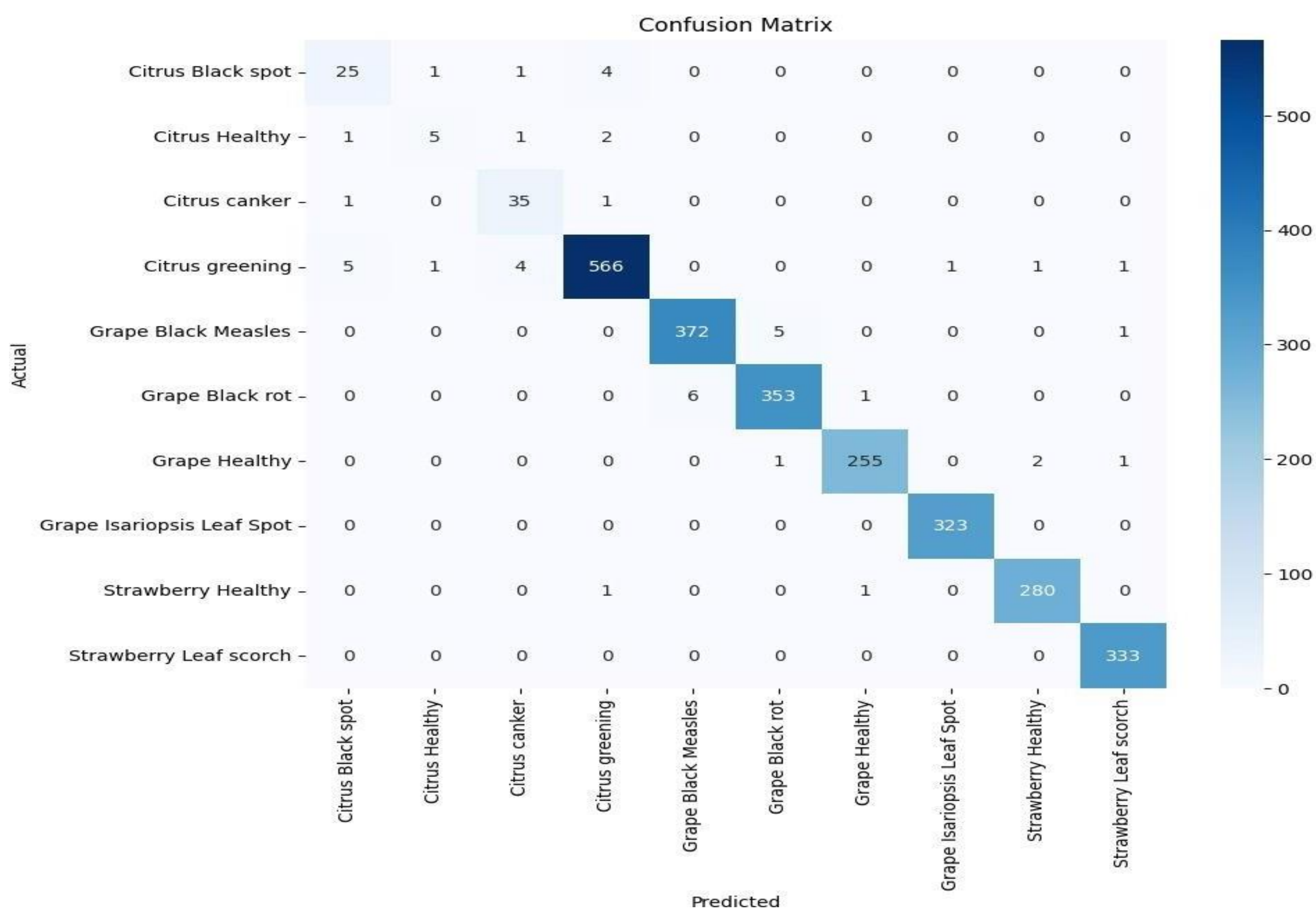

figure 4.4: Confusion Matrix of InceptionV3

4.2.3 industrial crops:

- **cotton**
- **Sugarcane**

IMAGES: 1,696

CLASSES: 3 {12 Disease - 2 healthy}

Model	Validation Accuracy	Test Accuracy
MobileNet	0.94112	0.92941

This confusion matrix represents the MobileNet model's classification performance on industrial crops including cotton and sugarcane. Despite the relatively small dataset size of 1,696 images (covering 12 diseases and 2 healthy classes), the model achieved a respectable test accuracy of 92.94%, indicating reliable detection across a complex set of conditions, as shown in the image below.

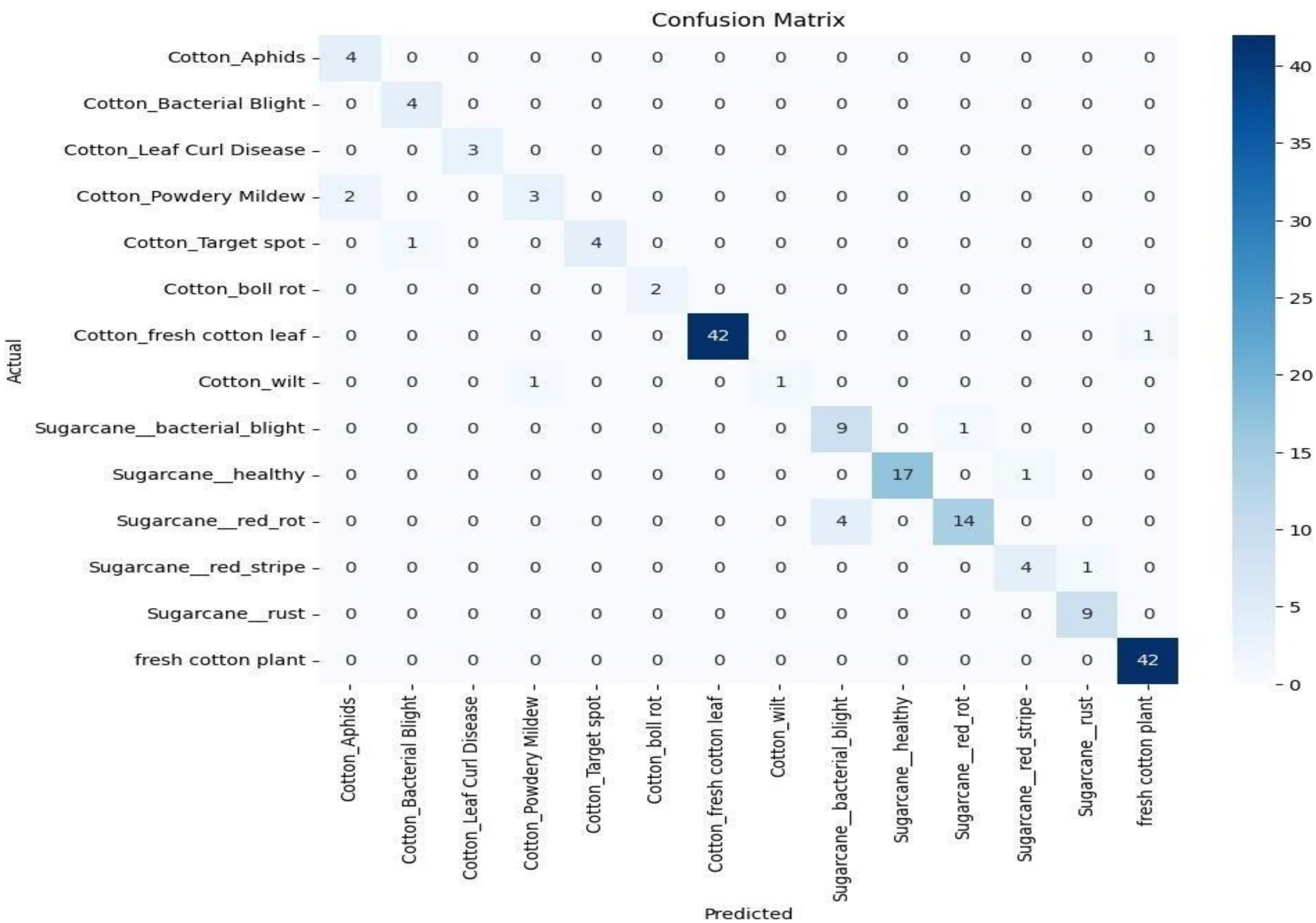

figure 4.5: Confusion Matrix of InceptionV3

4.2.4 Vegetables:

- **Potato**
- **Tomato**

IMAGES: 51,373

CLASSES: 3 {11 Disease - 2 healthy}

Model	Validation Accuracy	Test Accuracy
MobileNet	0.96231	0.95821

Shown here is the confusion matrix for vegetable crops, specifically potato and tomato, using the MobileNet model. With a large dataset of 51,373 images and 13 total classes (11 disease + 2 healthy), the model attained a test accuracy of 95.82%, highlighting its robustness in handling complex disease variability in vegetable plants, as shown in the image below.

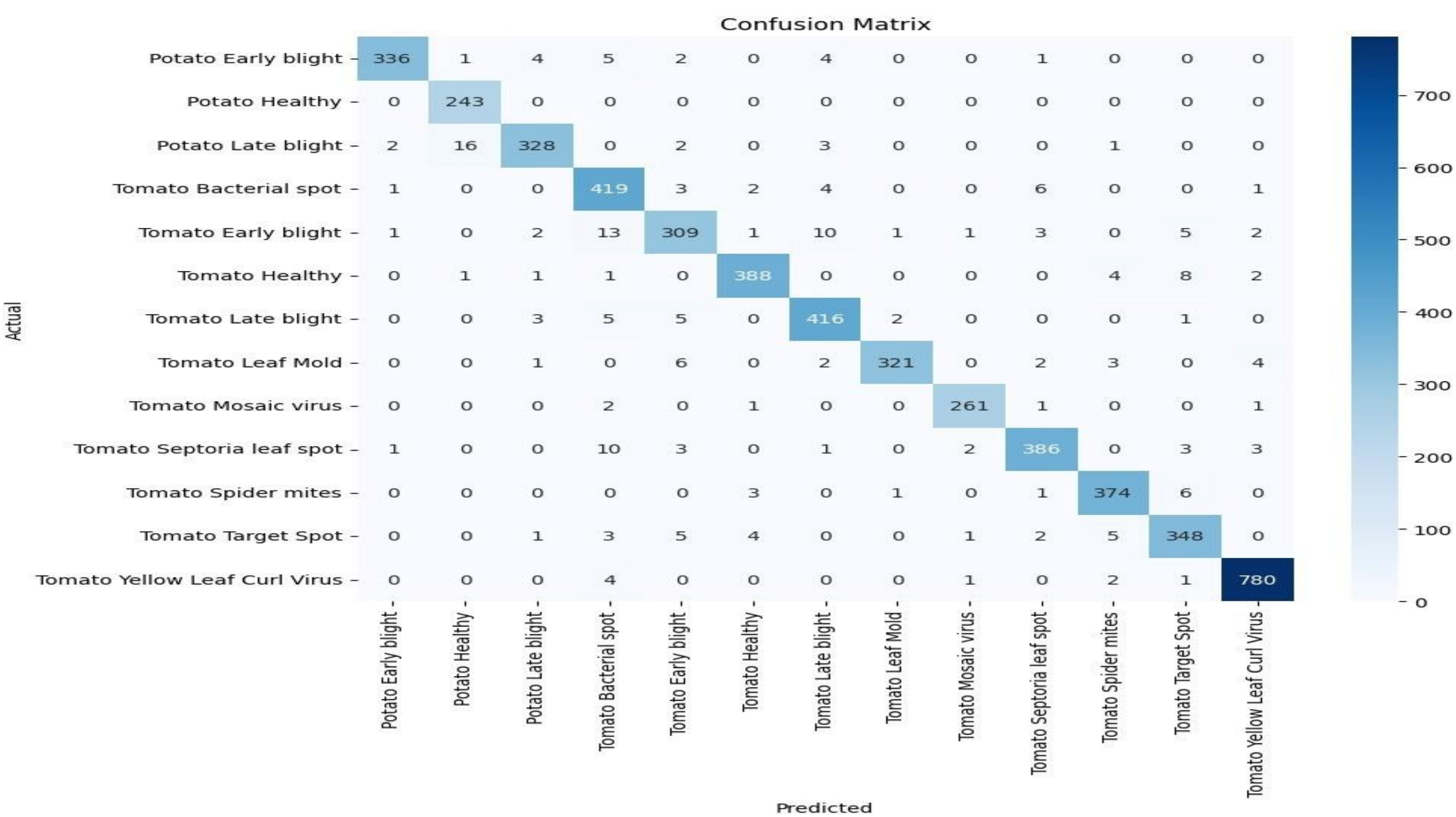

figure 4.6: Confusion Matrix of MobileNet

4.3. Results Explainable AI (XAI) with Grad-CAM:

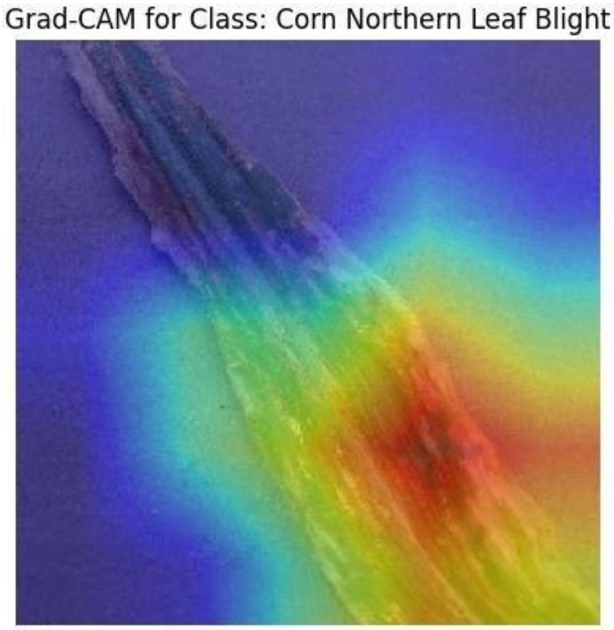

figure 4.7: XAI with Grad-CAM

figure 4.8: Confusion XAI with Grad-CAM

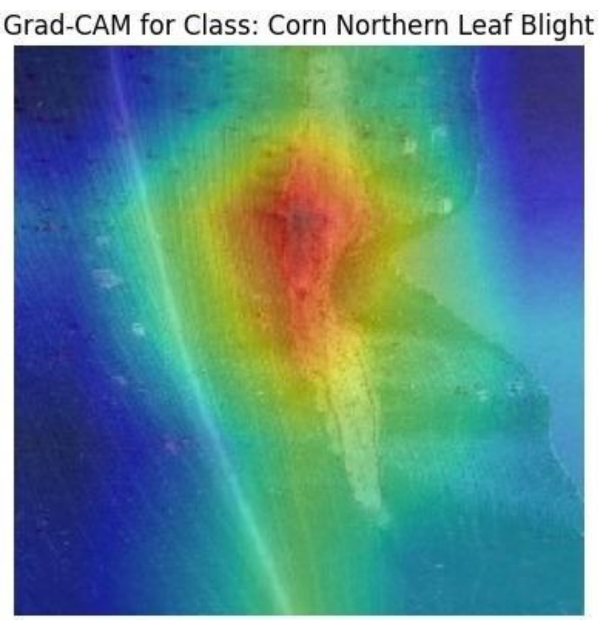

figure 4.9.: Confusion XAI with Grad-CAM

figure 4.10: Confusion XAI with Grad-CAM

4.4. Prototyping:

4.4.1 Mobile App

figure 4.11: Open App

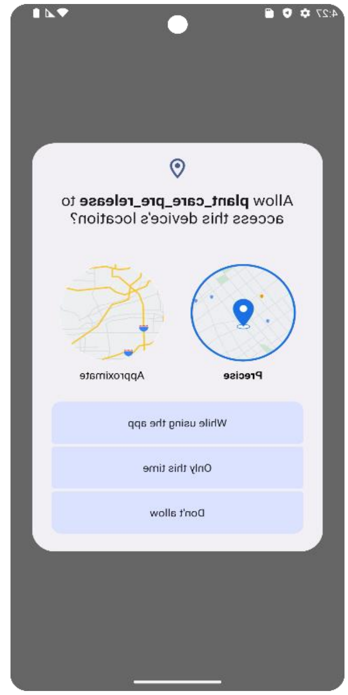

figure 4.12:Location Permissions

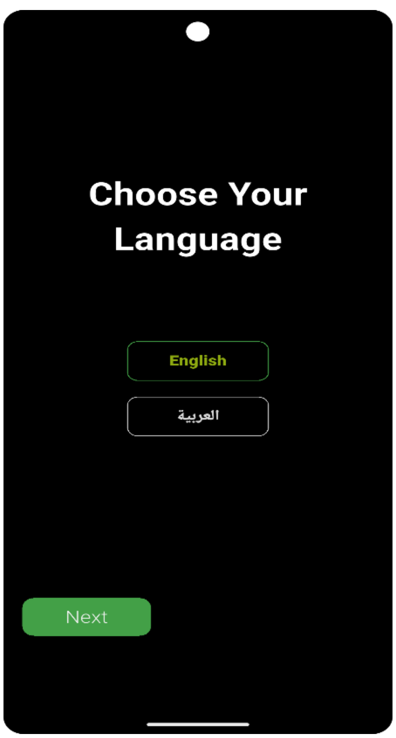

figure 4.13:Language Selection

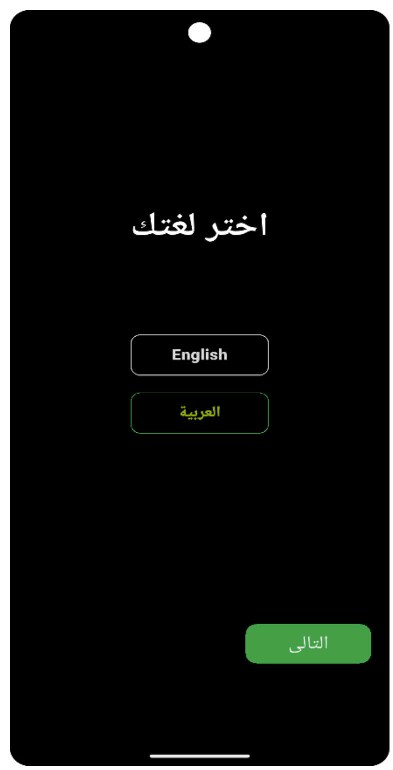

figure 4.14:Language Selections

figure 4.15:Onboarding screen1

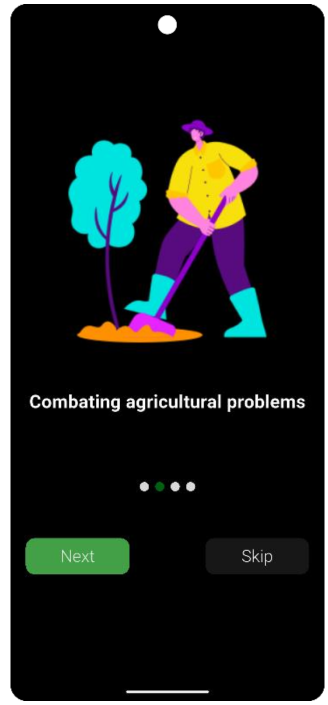

figure 4.16:Onboarding Screen 2

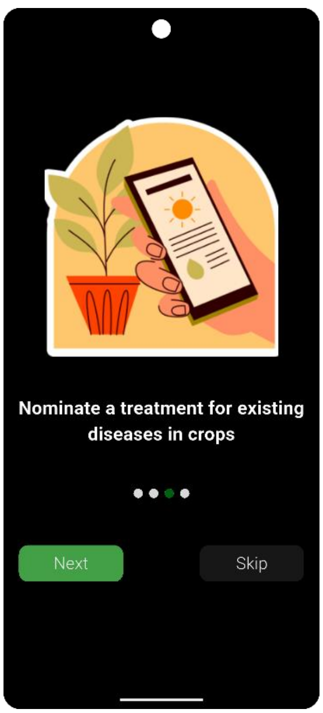

figure 4.17:Onboarding screen3

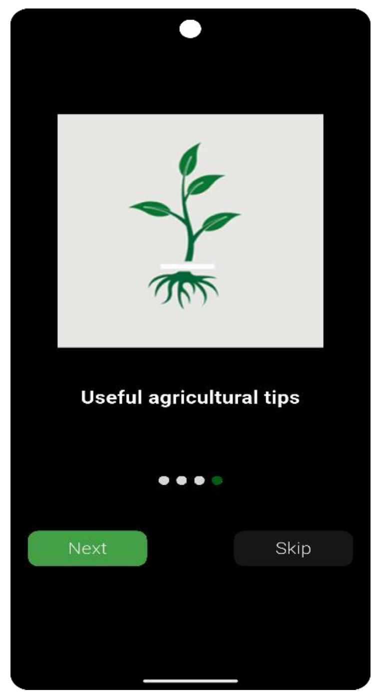

figure 4.18:Onboarding screen4

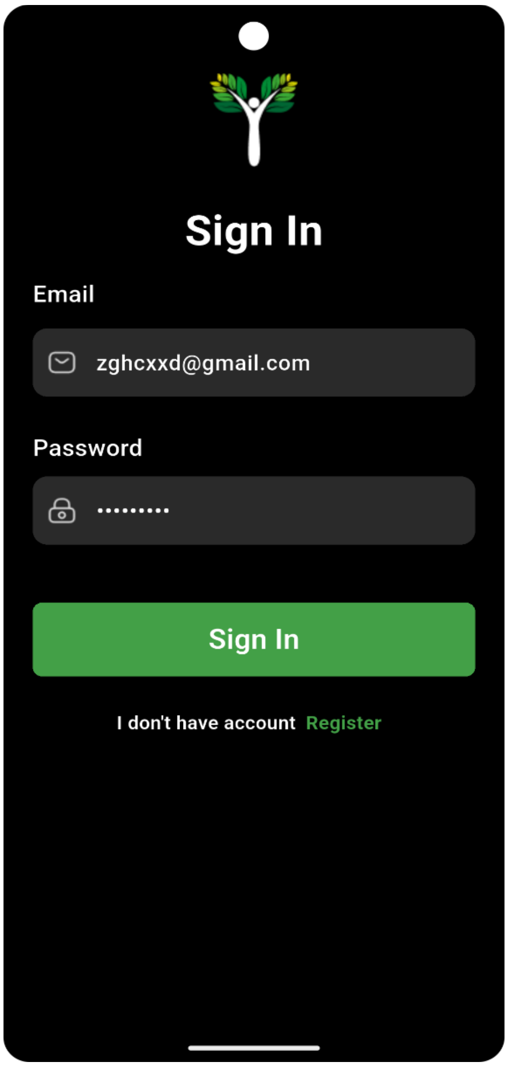

figure 4.19:Sign in Screen

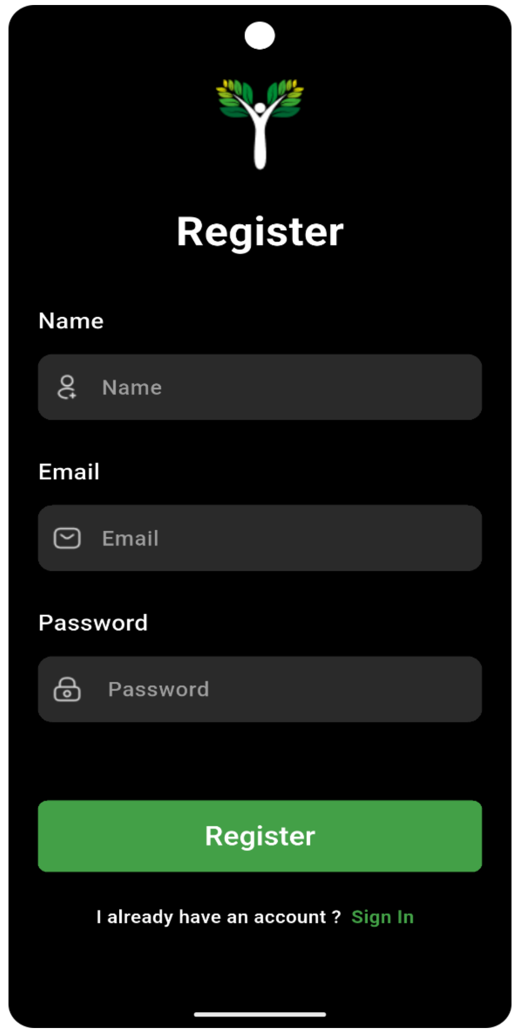

figure 4.20: Sign up Screen

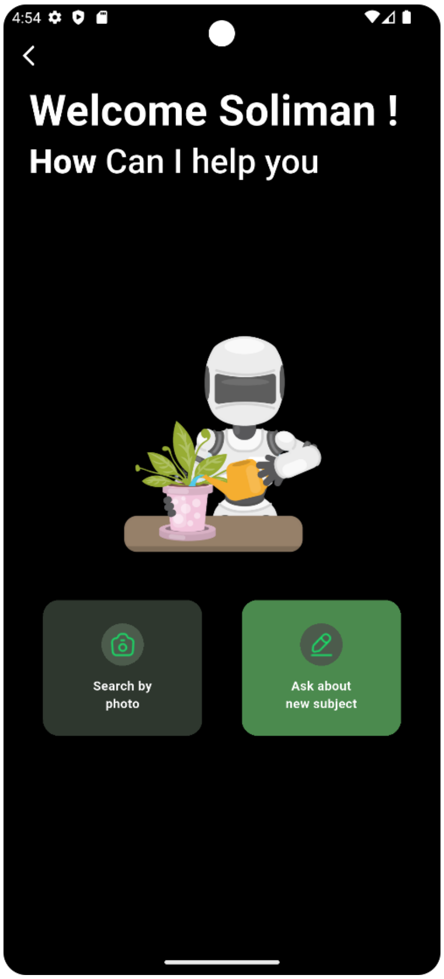

figure 4.21:Chatbot Screen

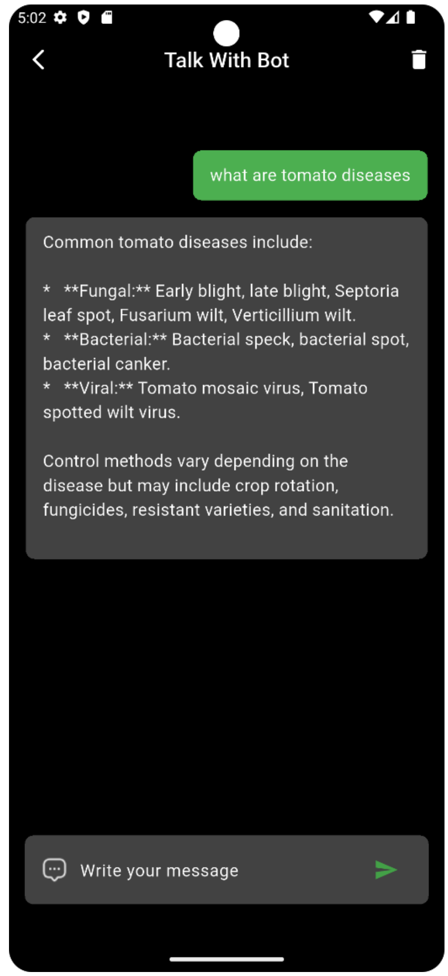

figure 4.22:Chat Screen

figure 4.23:Home Screen

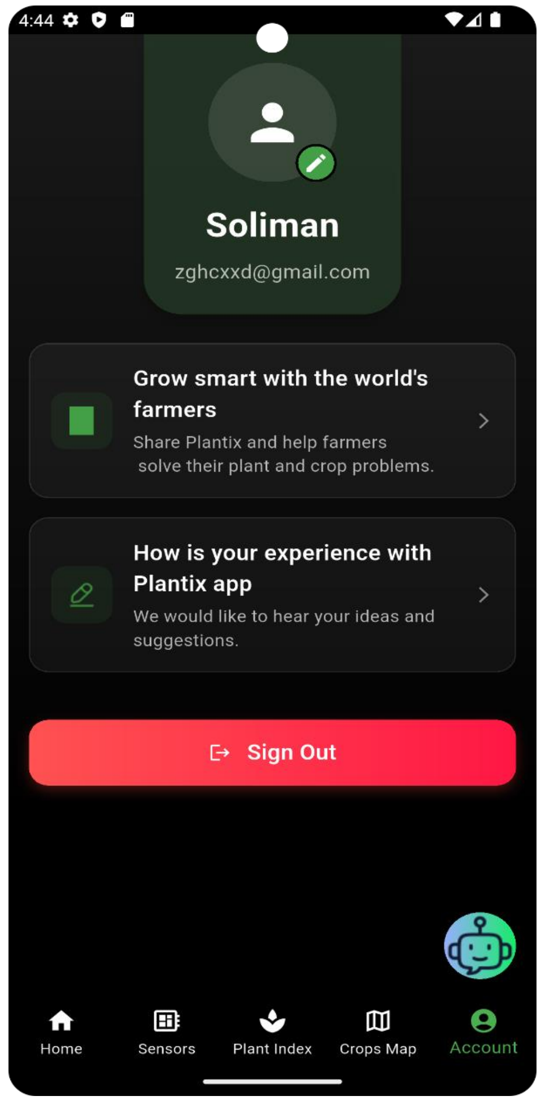

figure 4.24:Profile Screen

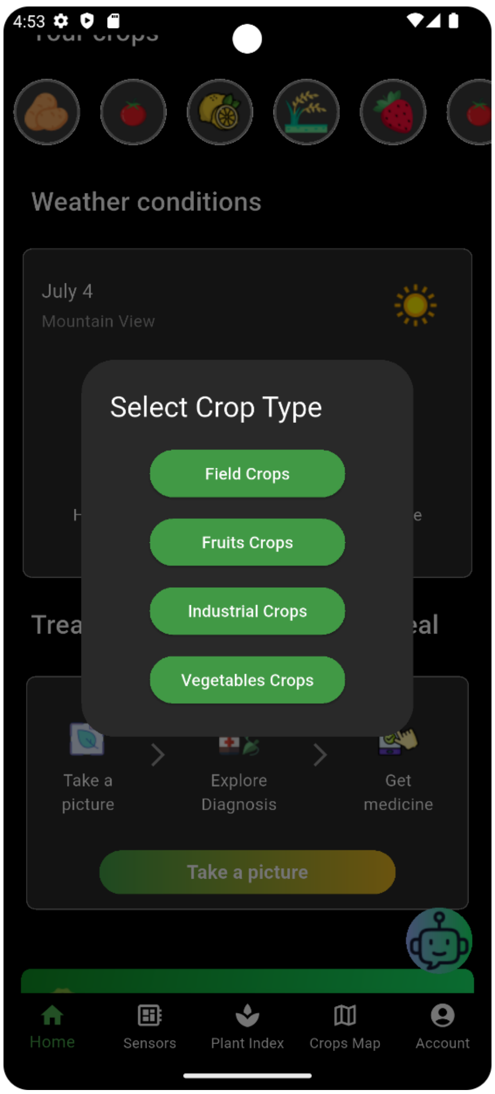

figure 4.25:Crop Selection

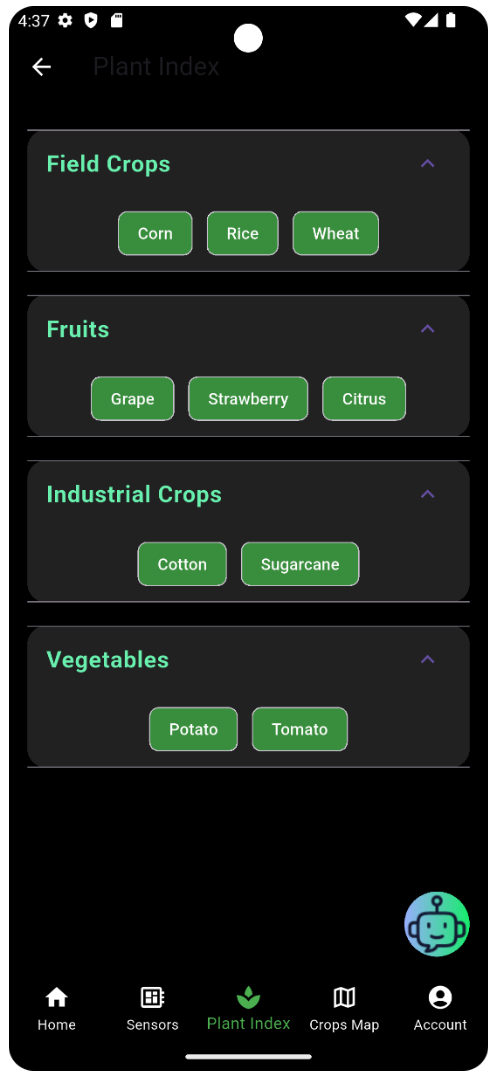

figure 4.26: Plant Index

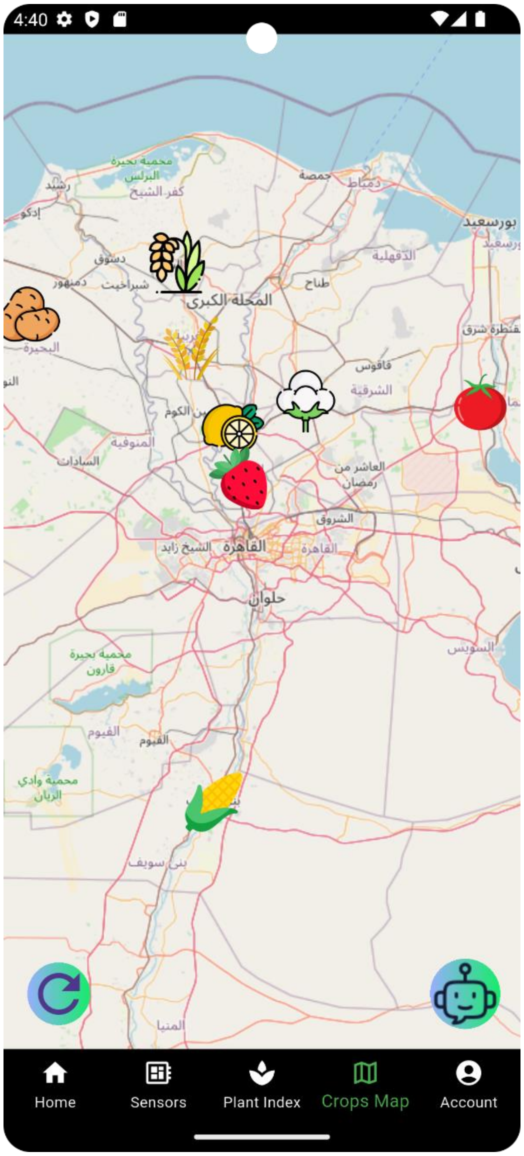

figure 4.27: Crops Map

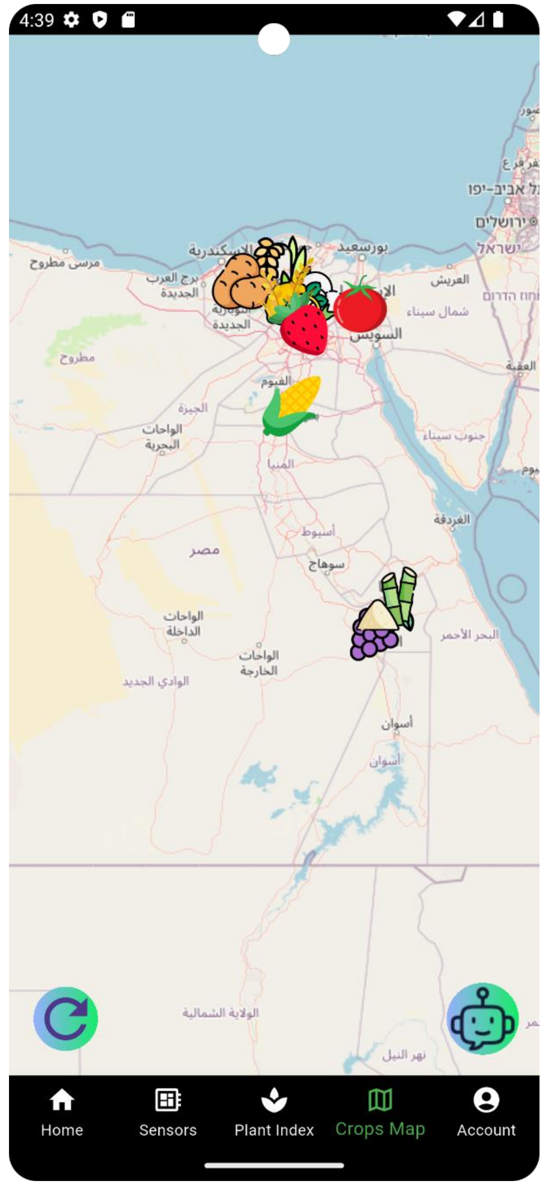

figure 4.28:Crops Map

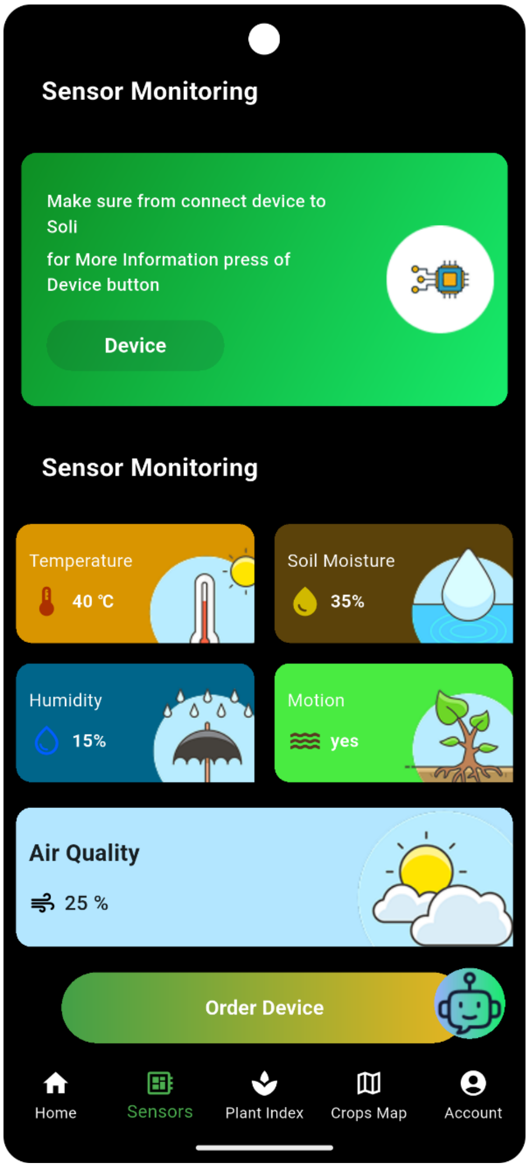

figure 4.29:Sensors Page

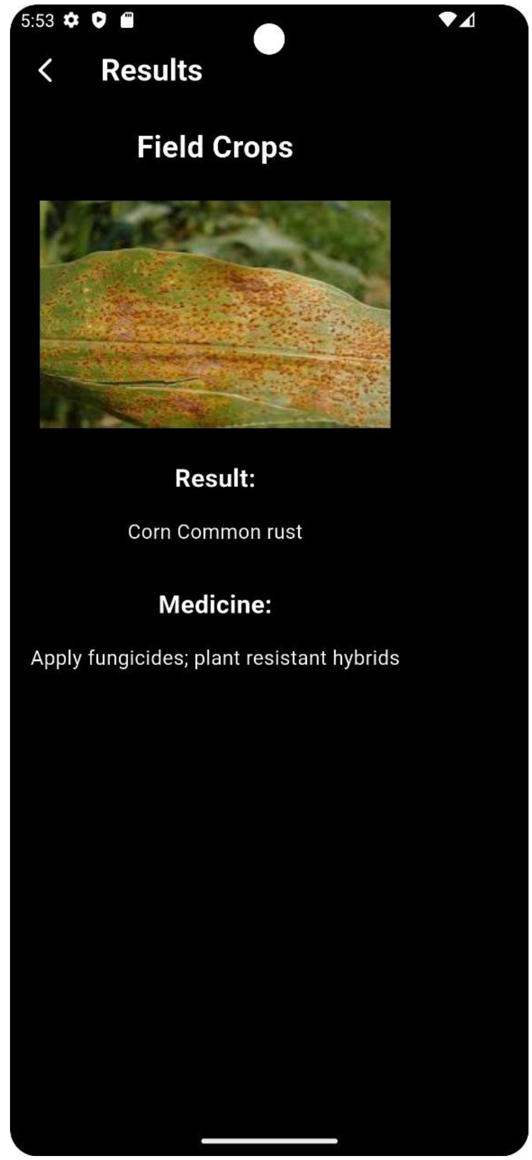

figure 4.30:Result Page

5.1 Website Dashboard

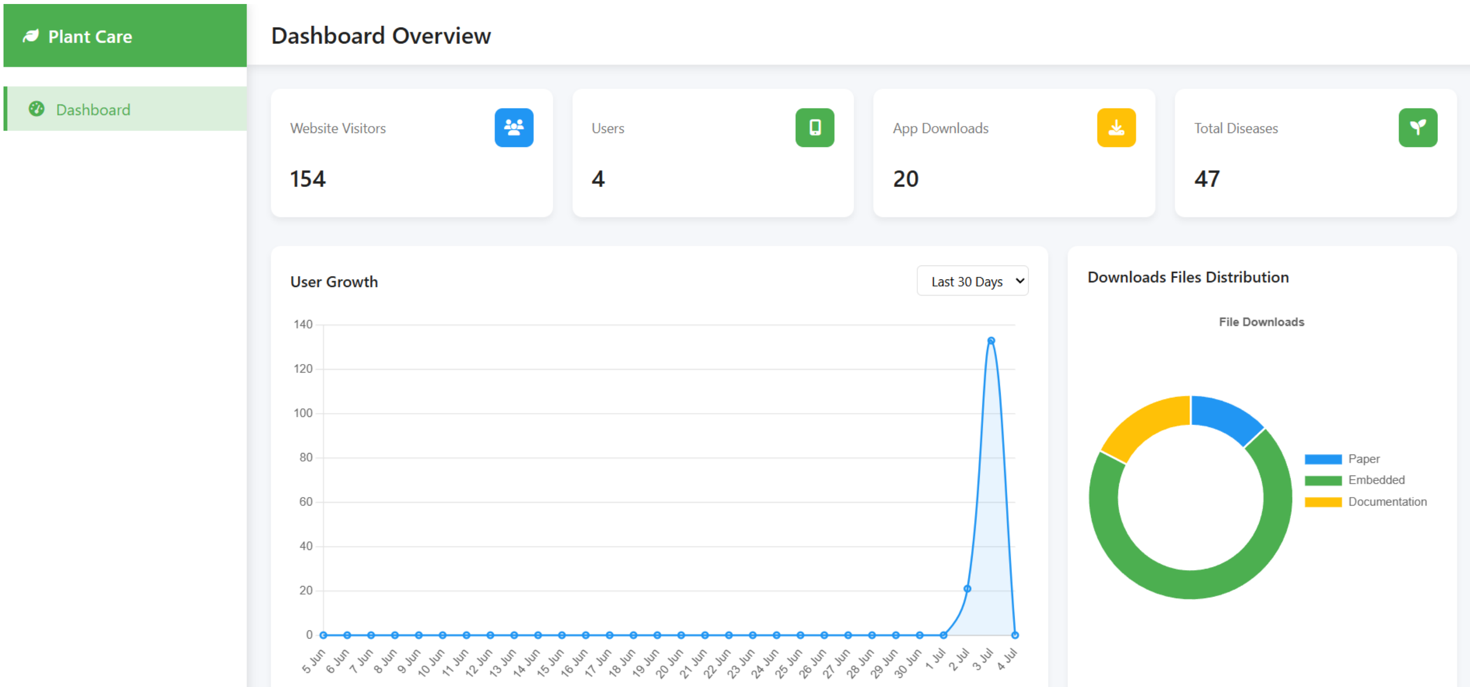

figure 5.1: Dashboard Overview

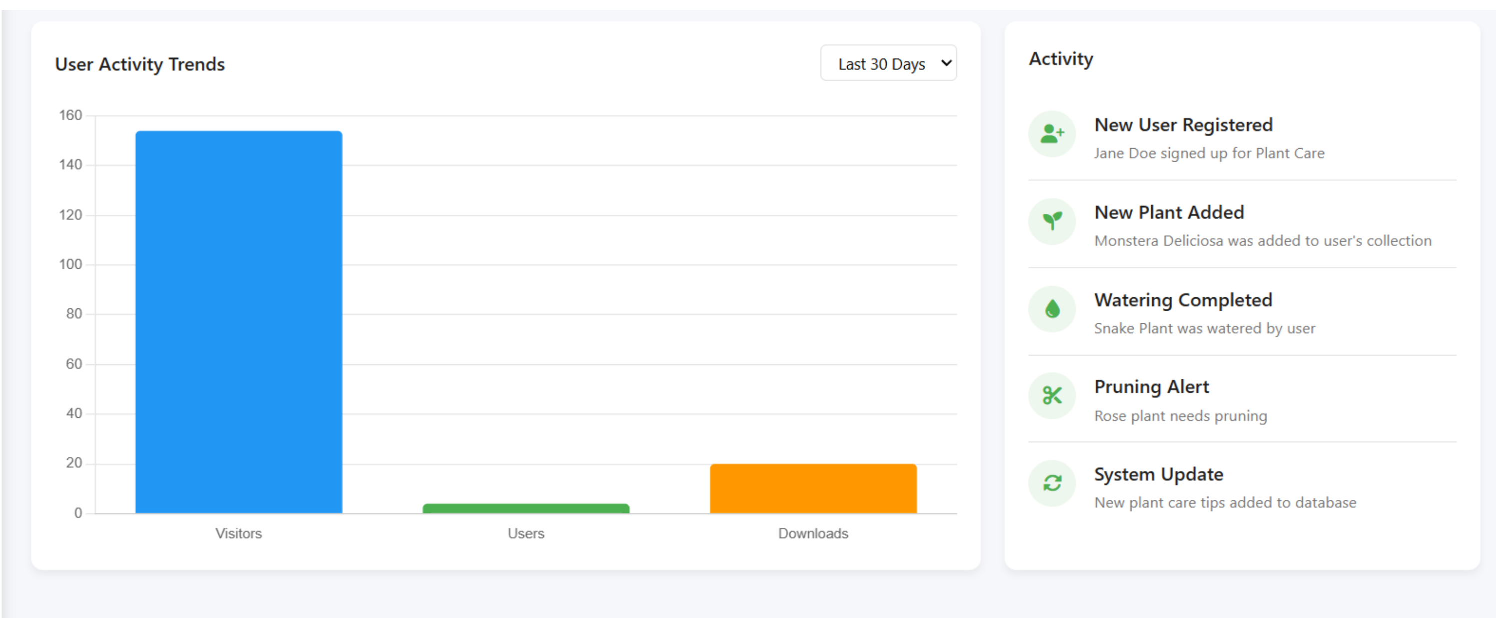

figure 5.2: Dashboard Overview

5.1.1 Environmental Sensors Simulation

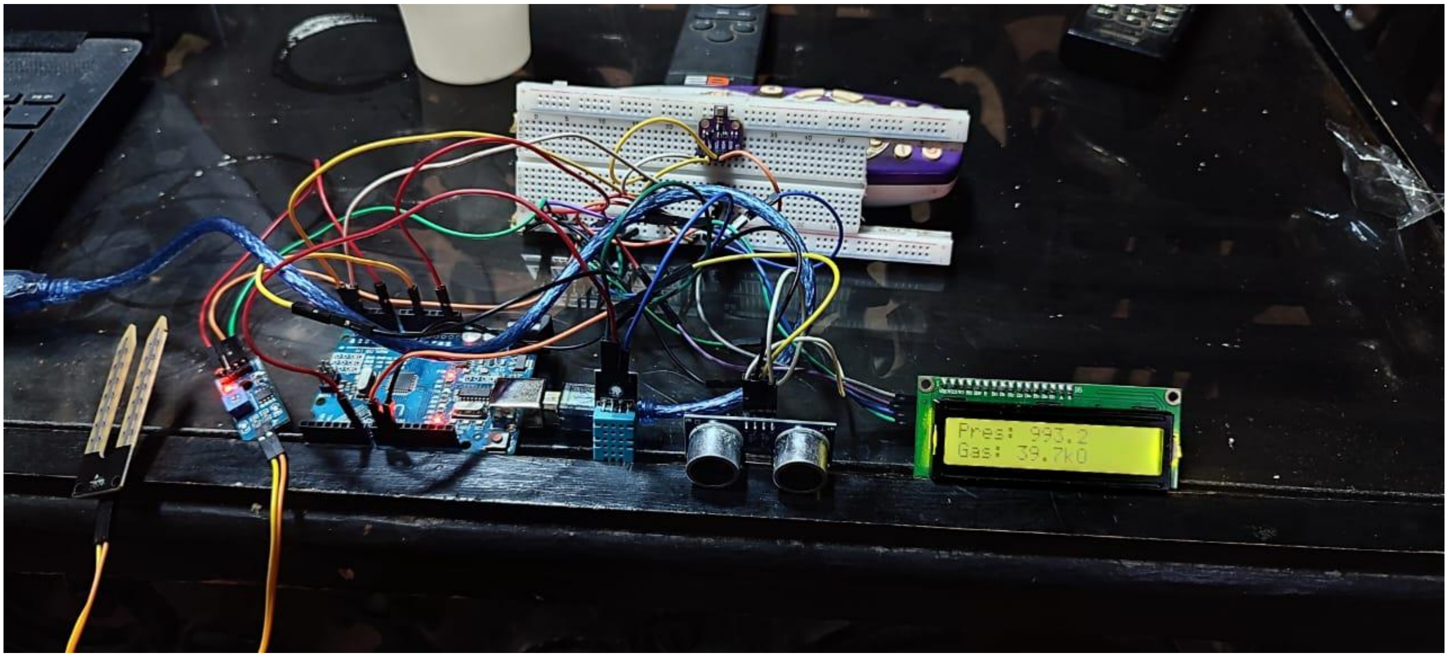

figure 5.3: Environmental sensor

Components:

1. **Arduino UNO**

Arduino controls all sensors

figure 5.4: Arduino UNO

2. Ultrasonic

Ultrasonic measures plant distance

figure 5.5: Ultrasonic

2. **motion sensor**

4. DHT11

Motion sensor detects movement

figure 5.6: Motion Sensor

DHT11 reads environmental data

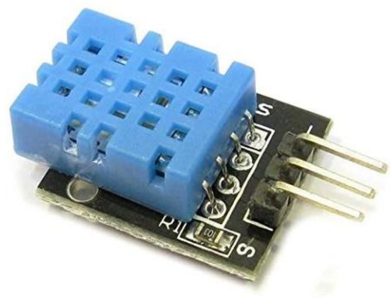

figure 5.7: DHT11

5. BME 680

BME680 senses air quality

figure 5.8: BME 680

6. Soil Moisture Sensor

Moisture sensor checks soil

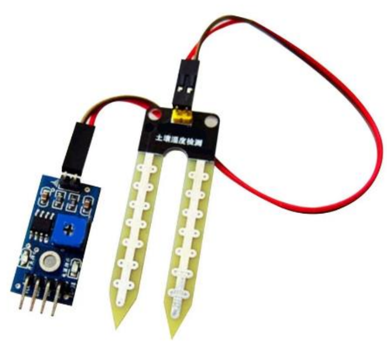

figure 5.9: Soil Moisture Sensor

7. LCD 2x16 I2C Display

LCD displays live readings

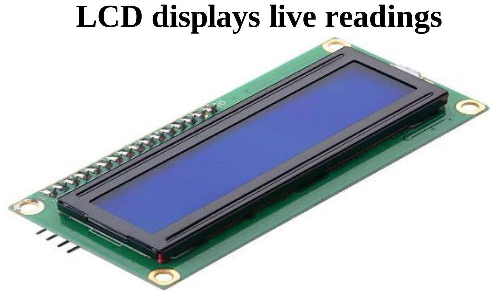

figure 5.10: LCD 2x16 I2C Display

C h a p t e r   F i v e

CONCLUSIONS AND FUTURE WORK

This project has explored the development of a plant disease detection system that leverages the power of deep learning, especially Convolutional Neural Networks, integrated with explainable AI techniques to enhance transparency and trust. The main objective was to handle the disadvantages of traditional plant disease diagnosis methods, which usually have been slow, time-consuming, labor-intensive, and full of errors. It intends to automate this process by developing an efficient, reliable approach that can be adopted for real-time monitoring of the plant's health.

A proper literature review in this paper summarized previous research carried out on various deep learning models including VGG-16, MobileNet, as well as variations of CNN. This review has done a good job in pointing out the strengths and shortcomings of these models; high accuracy that may be achieved with their limitations, such as the "black box" nature of decision-making and dependence on data quality. These insights motivated us to focus on the incorporation of XAI techniques such as LIME and SHAP to make the predictions more interpretable for users, thus increasing user trust and adoption in the agricultural sector.

The most critical events that took place during the development process involved planning and data collection. The other significant milestones, in order of occurrence, are UI/UX design, data preprocessing, development of a mobile application, and the creation of a dashboard-all contributing to the realization of this project. The prototype, a mobile application, and dashboard designed at first ensure ease in usability for farmers. This enables them to easily upload images of their crops for disease diagnosis and access important data and information on plant health. The sensor system developed complements the project by gathering real-time environmental data, further enhancing its utility.

The research presented here therefore underlines the potential of using deep learning and XAI in practical agricultural applications. Our system allows automatic disease detection with explainable AI to enable farmers to raise their yield by reducing crop loss, thus improving food security.

Future Work

- Further Research: In the future, we would like to extend the work that was undertaken by the project through further research. Some of the areas where further research could be useful are as follows:

- Model Improvements: The application of different deep learning models together may achieve even better results. Ensemble models are such models, while models like the Vision Transformers can also offer new insights and improvements.

- Diversification of Data: Increasing the size and diversity, including more images for a wide variety of crops and diseases taken under varying environmental conditions, would substantially improve the generalization and robustness of the model. In this regard, too, consideration of active learning techniques will be wise to consider.

- More Advanced XAI Techniques: Studying more XAI techniques will facilitate drawing insights in-depth and make the models more transparent and explainable. Combining different XAI techniques such as LIME and SHAP will enable more thorough feature importance.

- Sensor Integration: The sensor system can be extended to encompass a greater number of parameters, like nutrient levels and soil analysis, in order to further expand the holistic approach toward plant monitoring. This may also allow for more accurate insights by integrating the data streams into the dashboard.

- Deployment: Expanding the deployment infrastructure to ensure the system can be used in more diverse environments would help in bringing the work to a wider audience. This also includes exploring cloud based solutions.

REFERENCES

[1] AMRITAVARSHINI, S., et al. Image-Based Plant Leaf Disease Detection using Deep Learning. In: 2024 International Conference on Signal Processing, Computation, Electronics, Power and Telecommunication (IConSCEPT). IEEE, 2024. p. 1-6

[2] Rauf, S. (2024). Insecticide resistance in agricultural pests: challenges and solutions. Trends Biotech Plant Sci, 2(1), 36-41.

[3] Nigar, N., Faisal, H. M., Umer, M., Oki, O., & Lukose, J. (2024). Improving Plant Disease Classification with Deep Learning based Prediction Model using Explainable Artificial Intelligence. IEEE Access.

[4] Ajakwe, S. O., Esomonu, N. F., Deji-Oloruntoba, O., Ajakwe, I. U., Lee, J. M., & Kim, D. S. (2024). Machine Learning in UAV-Assisted Smart Farming. In Applications of Machine Learning in UAV Networks (pp. 217-245). IGI Global.

[5] Kaif, A. D., Alam, K. S., & Das, S. K. (2024). Blockchain based sustainable energy transition of a Virtual Power Plant: Conceptual framework design & experimental implementation. Energy Reports, 11, 261-275.

[6] Savary, S., et al. (2019). The global burden of pathogens and pests on major food crops. Nature Ecology & Evolution, 3(3), 430-439.

[7] Pacal, I., Kunduracioglu, I., Alma, M. H., Deveci, M., Kadry, S., Nedoma, J., ... & Martinek, R. (2024). A systematic review of deep learning techniques for plant diseases. Artificial Intelligence Review, 57(11), 304.

[8] Gaspar, D., Silva, P., & Silva, C. (2024). Explainable AI for Intrusion Detection Systems: LIME and SHAP Applicability on Multi-Layer Perceptron. IEEE Access.

[9] ŞAHiN, E., Arslan, N. N., & Özdemir, D. (2024). Unlocking the black box: an in-depth review on interpretability, explainability, and reliability in deep learning. Neural Computing and Applications, 1- 107.

[10] Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. Computers and Electronics in Agriculture, 145, 311-318.

[11] Benaissa, A. (2024). Rhizosphere: Role of bacteria to manage plant diseases and sustainable agriculture—A review. Journal of Basic Microbiology, 64(3), 2300361.

[12] Gupta, S., & Tripathi, A. K. (2024). Fruit and vegetable disease detection and classification: Recent trends, challenges, and future opportunities. Engineering Applications of Artificial Intelligence, 133, 108260.

[13] Archana, R., & Jeevaraj, P. E. (2024). Deep learning models for digital image processing: a review. Artificial Intelligence Review, 57(1), 11.

[14] Huo, D., Malik, A. W., Ravana, S. D., Rahman, A. U., & Ahmedy, I. (2024). Mapping smart farming: Addressing agricultural challenges in data-driven era. Renewable and Sustainable Energy Reviews, 189, 113858.

[15] Hassija, V., Chamola, V., Mahapatra, A., Singal, A., Goel, D., Huang, K., ... & Hussain, A. (2024). Interpreting black-box modelsa review on explainable artificial intelligence. Cognitive Computation, 1: 6(1), 45-74.

[16] Nabil, H. R., Mamun, R., Nasir, T., Netu, K. N. S., Bhowmik, A., & Karmaker, D. Harnessing Deep Learning for Plant Disease Analysis: Current Trends, Challenges, and Future Prospects. Challenges, and Future Prospects.

[17] Saleem, A., Anwar, S., Nawaz, T., Fahad, S., Saud, S., Ur Rahman, T., ... & Nawaz, T. (2024). Securing a sustainable future: the climate change threat to agriculture, food security, and sustainable development goals. Journal of Umm Al-Qura University for Applied Sciences, 1-17.

[18] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in plant science, 7, 1419.

[19] Jung, M., Song, J. S., Shin, A. Y., Choi, B., Go, S., Kwon, S. Y., ... & Kim, Y. M. (2023). Construction of deep learning-based disease detection model in plants. Scientific reports, 13(1), 7331.

[20] Sultana, N., Shorif, S. B., Akter, M., & Uddin, M. S. (2023). A dataset for successful recognition of cucumber diseases. Data in Brief, 49, 109320.

[21] Mallick, M. T., Biswas, S., Das, A. K., Saha, H. N., Chakrabarti, A., & Deb, N. (2023). Deep learning based automated disease detection and pest classification in Indian mung bean. Multimedia Tools and Applications, 82(8), 12017-12041.

[22] Nigar, N., Faisal, H. M., Umer, M., Oki, O., & Lukose, J. (2024). Improving Plant Disease Classification with Deep Learning based Prediction Model using Explainable Artificial Intelligence. IEEE Access.

[23] de Benito Fernández, M., Martínez, D. L., González-Briones, A., Chamoso, P., & Corchado, E. S. (2023, June). Evaluation of XAI Models for Interpretation of Deep Learning Techniques' Results in Automated Plant Disease Diagnosis. In Sustainable Smart Cities and Territories International Conference (pp. 417-428). Cham: Springer Nature Switzerland.

[24] New Plant Diseases Dataset (https://www.kaggle.com/datasets/vipoooool/new-plant-diseases-dataset)

[25] Huo, D., Malik, A. W., Ravana, S. D., Rahman, A. U., & Ahmedy, I. (2024). Mapping smart farming: Addressing agricultural challenges in data-driven era. Renewable and Sustainable Energy Reviews, 189, 113858.

[26] Plant Village for Object Detection (https://www.kaggle.com/datasets/sebastianpalaciob/plantvillage-for-object-detection-yolo)

[27] PlantifyDr Dataset (https://www.kaggle.com/datasets/lavaman151/plantifydr-dataset)

[28] Plant Disease Classification Merged Dataset (https://www.kaggle.com/datasets/alinedobrovsky/plant-disease-classification-merged-dataset)

[29] Plant Disease Expert Dataset (https://www.kaggle.com/datasets/sadmansakibmahi/plant-disease- expert)

[30] Mango Leaf Diesasse Dataset (https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease- dataset)